Fusion of Q-system defects and juxtaposition of boundary conditions

Notes towards a proof of a remark of Bischoff–Kawahigashi–Longo–Rehren

Notes prepared for A. Bailey

18 October 2026

Abstract

These notes give a proof, modulo standard facts about tensor categories that are listed explicitly, of a remark at the end of Section 5 of Bischoff–Kawahigashi–Longo–Rehren: that the juxtaposition of boundaries is described by the braided product of Q-systems, and the juxtaposition of boundary conditions $\mathbf{m}_1 \colon \mathbf{A} \to \mathbf{B}$, $\mathbf{m}_2 \colon \mathbf{B} \to \mathbf{C}$ by $\mathbf{m}_1 \otimes_{\mathbf{B}} \mathbf{m}_2$. Boundaries are realised as the Q-system defects of Bartels–Douglas–Henriques and juxtaposition as Connes fusion of defects. The first half is Theorem 1.2; the second is Theorem 9.7, which is the capstone of the notes.

Contents

1 Introduction **2**
- 1.1 The statement . . . 2
- 1.2 Main results . . . 3
- 1.3 What is proved and what is quoted . . . 3
- 1.4 Plan . . . 3

2 Background I: nets, defects, Q-systems **3**
- 2.1 Nets and the vacuum sector . . . 4
- 2.2 Defects . . . 4
- 2.3 Q-systems and their defects . . . 5
- 2.4 Connes fusion and fusion of defects . . . 5
- 2.5 Braided products of Q-systems . . . 6
- 2.6 Conventions and standing hypotheses . . . 6
- 2.7 Standard facts from the literature (cited without proof) . . . 7

3 Background II: tensor categories **7**
- 3.1 Setting . . . 7
- 3.2 Quoted facts . . . 7
- 3.3 Q-systems as algebras . . . 8
- 3.4 The projection formula . . . 8
- 3.5 Two consequences of the quoted facts . . . 9

4 Algebraic lemmas on a Q-system **10**

5 The fused space **11**

6 Identification with the braided product **13**
6.1 Vacuum gluing . . . 13
6.2 The net-level theorem . . . 14
6.3 Sign of the braided product . . . 14
6.4 Change of chart; the fused object is a defect . . . 14

7 Consequences of Theorem 1.2 **15**
7.1 Irreducible and reducible products . . . 15
7.2 Full centres and phase boundaries . . . 16
7.3 Corners . . . 17
7.4 Associativity . . . 18

8 Products of full centres **18**

9 Proof of Remark 1.1 **19**
9.1 (E1) . . . 19
9.2 Three lemmas . . . 19
9.3 The summand over the phase $\mathbf{B}$ . . . 21
9.4 (E2) . . . 22
9.5 Interpretation . . . 23
9.6 What remains to be done . . . 23

10 Applications **23**

A Numerical checks **23**

1 Introduction

1.1 The statement

Let A be a completely rational conformal net with modular category $\mathcal{C} = \mathrm{DHR}(A)$, and $\mathcal{A} = A \otimes \bar{A}$. For a simple Q-system $\mathbf{A}$ in $\mathcal{C}$ the full centre $Z[\mathbf{A}]$ is a commutative Q-system in $\mathrm{DHR}(\mathcal{A}) \supset \mathcal{C} \boxtimes \mathcal{C}^{\mathrm{opp}}$, and its extension of $\mathcal{A}$ is the two-dimensional local net of the phase $\mathbf{A}$. [BKLR] show that the braided product $Z[\mathbf{A}] \times Z[\mathbf{B}]$ decomposes, by its centre, into pieces labelled by the irreducible $\mathbf{A}$–$\mathbf{B}$-bimodules $\mathbf{m}$, the *boundary conditions* between the phases $\mathbf{A}$ and $\mathbf{B}$. At the end of their Section 5 they remark:

Remark 1.1 ([BKLR, end of §5]). **(E1)** The juxtaposition of two boundaries is described by the (associative) braided product of Q-systems. **(E2)** The juxtaposition of boundary conditions $\mathbf{m}_1 \colon \mathbf{A} \to \mathbf{B}$ and $\mathbf{m}_2 \colon \mathbf{B} \to \mathbf{C}$ is described by $\mathbf{m}_1 \otimes_{\mathbf{B}} \mathbf{m}_2$.

To make this a mathematical statement one needs a notion of boundary and of juxtaposition. We use those of [BDH]: a Q-system Q in $\mathcal{A}$ determines a defect D_Q from $\mathcal{A}$ to $\mathcal{A}$, and defects are composed by Connes fusion $\circledast$. The boundary condition $\mathbf{m}$ becomes the irreducible defect $D_{\mathbf{m}}$ cut out of $D_{Z[\mathbf{A}]} \circledast D_{Z[\mathbf{B}]}$ by the central projection labelled $\mathbf{m}$.

1.2 Main results

Theorem 1.2 ((E1); Theorem 6.4, Proposition 7.7)**.** *For Q-systems Q, Q' in $\mathcal{A}$ localized as in Assumption 2.13 there is a Möbius map h with $D_Q \circledast D_{Q'} \cong D_{Q^h \times Q'}$, where Q^h is equivalent to Q and $\times$ is the braided product (sign fixed by the orientation convention C1). The fused object is a defect without any finite-index assumption, $H_0(D_Q) \boxtimes H_0(D_{Q'}) \cong L^2(D_{Q^h \times Q'}(I))$, and fusion of Q-system defects is associative.*

Theorem 1.3 ((E2); Theorem 9.7)**.** *For boundary conditions $\mathbf{m}_1 \colon \mathbf{A} \to \mathbf{B}$, $\mathbf{m}_2 \colon \mathbf{B} \to \mathbf{C}$,*

$$D_{\mathbf{m}_1} \circledast D_{\mathbf{m}_2} \cong \bigoplus_{\mathbf{m}' \in \mathrm{Irr}({}_{\mathbf{B}}\mathcal{C}_{\mathbf{B}})} D_{(\mathbf{m}_1, \mathbf{m}', \mathbf{m}_2)},$$

a sum of irreducible defects with relative commutants $\mathrm{End}_{\mathbf{A}\text{-}\mathbf{C}}(\mathbf{m}_1 \otimes_{\mathbf{B}} \mathbf{m}' \otimes_{\mathbf{B}} \mathbf{m}_2)$. The summand $\mathbf{m}' = \mathbf{B}$ is canonically the summand $D_{(\mathbf{m}_1, \mathbf{m}_2)}$ of the three-fold fusion $D_{Z[\mathbf{A}]} \circledast D_{Z[\mathbf{B}]} \circledast D_{Z[\mathbf{C}]}$, which is non-zero, irreducible, and has relative commutant $\mathrm{End}_{\mathbf{A}\text{-}\mathbf{C}}(\mathbf{m}_1 \otimes_{\mathbf{B}} \mathbf{m}_2) \cong \bigoplus_{\mathbf{m}} M_{N^{\mathbf{m}}_{\mathbf{m}_1 \mathbf{m}_2}}(\mathbb{C})$; and the $D_{(\mathbf{m}_1, \mathbf{m}_2)}$ are exactly the irreducible summands of the three-fold fusion.

Thus "juxtaposition through the phase $\mathbf{B}$" is described by $\mathbf{m}_1 \otimes_{\mathbf{B}} \mathbf{m}_2$, as the remark expects; fusion over the ambient net $\mathcal{A}$ in addition allows any $\mathbf{B}$–$\mathbf{B}$ boundary condition $\mathbf{m}'$ in the middle.

1.3 What is proved and what is quoted

Everything about nets is proved here, using the axioms of [BDH] and the facts L1–L3 of Section 2.7. On the categorical side we use [BKLR] and a list of standard facts about module categories and centres, collected as (T1)–(T7) in Section 3; each is either a definition-level property or a theorem in the literature cited there. One further input, Proposition 3.4, is proved here at the level of objects and for the unit, but its compatibility with products in general is a diagram chase that is described and not drawn (Remark 3.5); it is used for the algebra structure of the relative commutants and for irreducibility (Theorem 8.2), not for the existence, non-vanishing and canonical labelling of the summands (Theorem 9.5). All proofs are by hand and have not been machine-checked; Appendix A records numerical tests in group-theoretical models.

1.4 Plan

Section 2 recalls nets, defects, Q-systems, Connes fusion and braided products; Section 3 the tensor-category background. Sections 4–6 prove Theorem 1.2. Section 7 derives the consequences used later (direct sums, corners, associativity, the two-fold product of full centres). Section 8 computes the products of several full centres. Section 9 proves Remark 1.1. Applications to two-dimensional conformal QFT (the time-zero dictionary with the boundary nets of [BKLR2], fusion through an intermediate phase, hard boundaries, sectors, junctions, crossings, corners, lightlike boundaries, hard boundaries meeting at a point) are in the companion notes, Section 10.

2 Background I: nets, defects, Q-systems

References: [BDH] = *Conformal nets III*, arXiv:1310.8263; [BKLR] = the book of Bischoff–Kawahigashi–Longo–Rehren, arXiv:1407.4793. Statements are restated in the form used here.

2.1 Nets and the vacuum sector

A conformal net $\mathcal{A}$ assigns to each oriented interval I a von Neumann algebra $\mathcal{A}(I)$, covariantly, with injective (anti)homomorphisms for embeddings; it satisfies locality, strong additivity, Haag duality $\mathcal{A}(I)' = \mathcal{A}(I')$ on the vacuum space, the Reeh–Schlieder property, and *inner covariance*: if $\varphi \in \mathrm{Diff}_+(I)$ is the identity near ∂I then $\mathcal{A}(\varphi) = \mathrm{Ad}(u)$ for a unitary $u \in \mathcal{A}(I)$.

For a conformal circle S there is a vacuum space $H_0(S, \mathcal{A})$ and, for each $\varphi \in \mathrm{Conf}(S, S')$, an operator $H_0(\varphi, \mathcal{A})$, unitary if φ preserves orientation and anti-unitary otherwise, with

$$\mathcal{A}(\varphi)(a) = H_0(\varphi)\, a\, H_0(\varphi)^* \ (\varphi \in \mathrm{Conf}_+), \qquad \mathcal{A}(\varphi)(a) = H_0(\varphi)\, a^*\, H_0(\varphi)^* \ (\varphi \in \mathrm{Conf}_-). \quad \text{(F2)}$$

For each interval $I \subset S$ there is a canonical unitary $v_I \colon H_0(S, \mathcal{A}) \to L^2(\mathcal{A}(I))$, and the modular conjugation of $(\mathcal{A}(I), \Omega)$ is the anti-unitary of the reflection j_I fixing ∂I:

$$J_I = v_I\, H_0(j_I, \mathcal{A})\, v_I^*. \quad \text{(F1)}$$

(F1), (F2) are [BDH, Appendix, "The vacuum sector"]. The anti-unitaries are normalized by $H_0(\varphi)\Omega = \Omega$; this is a normalization, since $H_0(\varphi)$ is fixed by (F2) only up to a phase, $H_0(\varphi)\Omega$ is a vacuum vector, hence a phase times Ω, and rescaling by a phase does not change (F2) or $H_0(\varphi)^2 = 1$ for involutions (C3 below).

2.2 Defects

Definition 2.1 ([BDH])**.** A defect from $\mathcal{A}$ to $\mathcal{B}$ is a functor D from bicolored intervals to von Neumann algebras (separable preduals; morphisms are homomorphisms and anti-homomorphisms) which restricts to $\mathcal{A}$ on white and to $\mathcal{B}$ on black intervals, sends orientation-preserving embeddings to homomorphisms and orientation-reversing ones to anti-homomorphisms, and satisfies:

(1) *Isotony:* for genuinely bicolored $J \subset I$, $D(J) \to D(I)$ is injective.
(2) *Locality:* if $J, K \subset I$ have disjoint interiors, the images of $D(J)$, $D(K)$ in $D(I)$ commute.
(3) *Strong additivity:* if $I = J \cup K$, the images of $D(J)$ and $D(K)$ generate $D(I)$.
(4) *Vacuum sector:* let S be a bicolored circle, $I \subset S$ a genuinely bicolored interval, j a colour-preserving orientation-reversing involution of S fixing ∂I, $I' = j(I)$. Let $\alpha \colon D(I) \to \mathbf{B}(L^2 D(I))$ be the left action and $\beta \colon D(I') \xrightarrow{D(j)} D(I)^{\mathrm{op}} \to \mathbf{B}(L^2 D(I))$ the right action. If $J \subset I$ is a monochrome interval with $J \cap \partial I$ a single point and $\bar{J} = j(J)$, then $\alpha \otimes \beta$ on $D(J) \otimes_{\mathrm{alg}} D(\bar{J})$ extends to an action of $D(J \cup \bar{J})$.

D is irreducible if every $D(I)$, I genuinely bicolored, is a factor. For white or black I inside a genuinely bicolored J the map $D(I) \to D(J)$ need not be injective.

The vacuum sector of D is $H_0(D) = L^2(D(S^1_\dagger))$ for the standard bicolored circle, with left actions of all $D(I)$ for intervals $I \subset S^1$ avoiding the colour points in the manner of [BDH, Def. of $L^2(D)$].

Proposition 2.2 ([BDH], Haag duality for defects)**.** *(1) For a genuinely bicolored circle S and bicolored interval I, $D(I)$ and $D(I')$ are each other's commutants on $H_0(S, D)$. (2) If $J \in \mathrm{INT}_{\circ\bullet}$ and K is monochrome, $J, K \subset I \in \mathrm{INT}_{\circ\bullet}$, $J \cup K = I$ and $J \cap K$ is a point, then $D(J) = D(I) \cap D(K)'$.*

Proposition 2.3 ([BDH], inner covariance for defects)**.** *If I is genuinely bicolored and $\varphi \in \mathrm{Diff}_0(I)$ preserves the bicoloring and the local coordinate, then $D(\varphi)$ is an inner automorphism of $D(I)$.*

2.3 Q-systems and their defects

Definition 2.4. A Q-system in $\mathcal{A}$ is a triple (θ, w, x) where θ is a localized endomorphism of $\mathcal{A}([0,1])$ (the identity on $\mathcal{A}([0,\epsilon])$ and $\mathcal{A}([1-\epsilon,1])$ for some $\epsilon > 0$) and $w, x \in \mathcal{A}([0,1])$ satisfy, with a scalar d,

$$wa = \theta(a)w, \quad x\theta(a) = \theta^2(a)x \qquad (a \in \mathcal{A}([0,1])),$$
$$w^*x = \theta(w^*)x = 1, \quad xx = \theta(x)x, \quad xx^* = \theta(x^*)x, \quad w^*w = x^*x = d \cdot 1.$$

Two Q-systems (θ, w, x) and $(\mathrm{Ad}(u)\theta, uw, u\theta(u)xu^*)$, u unitary, are *equivalent*.

For a genuinely bicolored interval I put $I^\sharp := I_\circ \cup [0,1] \cup I_\bullet$ (the interval with a unit strip inserted at the colour point; this is I^+ in [BDH], renamed to avoid a clash with the fusion intervals below). Since θ is localized it has a unique extension $\theta^\sharp$ to $\mathcal{A}(I^\sharp)$ which is the identity on $\mathcal{A}(K)$ for $K \subset I^\sharp \setminus (0,1)$; we write θ for it.

Proposition 2.5 ([BDH, Prop. 1.27])**.** *Let $D_Q(I)$ be the algebra generated by $\mathcal{A}(I^\sharp)$ and an element v subject to*

$$va = \theta(a)v \ (a \in \mathcal{A}(I^\sharp)), \qquad v^* = w^*x^*v, \quad vv = xv, \quad w^*v = 1.$$

Then D_Q is an $\mathcal{A}$–$\mathcal{A}$ defect. Moreover $a \mapsto av$ is a bijection $\mathcal{A}(I^\sharp) \to D_Q(I)$, $E(av) := d^{-1}aw$ is a conditional expectation $D_Q(I) \to \mathcal{A}(I^\sharp)$ (faithful, Fact L1), $D_Q(K) = \mathcal{A}(K^\sharp) \vee \{v\}$ for genuinely bicolored $K \subset I$, and $d\,E(vv^*) = 1$.

Definition 2.6. The *weak identity* is the defect $\mathbf{1}^{\mathrm{w}}_{\mathcal{A}}\colon I \mapsto \mathcal{A}(I^\sharp) = D_{(\mathrm{id},1,1)}(I)$. The *identity defect* $\mathrm{Id}_{\mathcal{A}}$ is $I \mapsto \mathcal{A}(I)$. By [BDH, Example] $\mathrm{Id}_{\mathcal{A}} \ncong \mathbf{1}^{\mathrm{w}}_{\mathcal{A}}$ (the colour-point algebra $\bigcap_J \mathbf{1}^{\mathrm{w}}_{\mathcal{A}}(J) = \mathcal{A}([0,1])$ is non-trivial, that of $\mathrm{Id}_{\mathcal{A}}$ is $\mathbb{C}$), but they are equivalent via an invertible sector.

2.4 Connes fusion and fusion of defects

For a von Neumann algebra C, a right module H_1 and a left module H_2, $H_1 \boxtimes_C H_2$ is the completion of $\mathrm{Hom}_C(L^2C, H_1) \odot H_2$ with $\langle \xi \boxtimes \eta, \xi' \boxtimes \eta' \rangle = \langle \eta, R(\xi)^*R(\xi')\eta' \rangle$. Here $\xi \in H_1$ is *C-bounded* if $R(\xi)\colon L^2(C) \to H_1$, $\Omega_C \cdot c \mapsto \xi \cdot c$, is bounded, and $R(\xi)^*R(\xi') \in C$ acts on H_2. Vectors $\xi \boxtimes \eta$ with ξ bounded span a dense subspace, and $\|\xi \boxtimes \eta\| \le \|R(\xi)\|\|\eta\|$.

Definition 2.7 ([BDH])**.** For homomorphisms $A \leftarrow C^{\mathrm{op}}$, $C \to B$ and faithful modules ${}_AH$, ${}_BK$,

$$A \circledast_C B := (A \cap C^{\mathrm{op}\,\prime}) \vee (C' \cap B) \subset \mathbf{B}(H \boxtimes_C K).$$

Let I be a bicolored interval with the following charts: $I^+_\bullet = [0, \frac{1}{2}]$, $I^{++}_\bullet = [0, \frac{3}{2}]$, ${}^+I_\circ = [-\frac{1}{2}, 0]$, ${}^{++}I_\circ = [-\frac{3}{2}, 0]$ (the other halves unchanged), and $J = [0,1]$ embedded in $I^{++}_\bullet$ by $x \mapsto \frac{3}{2} - x$ and in ${}^{++}I_\circ$ by $x \mapsto x - \frac{3}{2}$. For defects ${}_{\mathcal{A}}D_{\mathcal{B}}$, ${}_{\mathcal{B}}E_{\mathcal{C}}$ and genuinely bicolored I,

$$(D \circledast_{\mathcal{B}} E)(I) := D(I^{++}) \circledast_{\mathcal{B}(J)} E({}^{++}I) = D(I^+) \vee E({}^+I),$$

the last equality by Haag duality (Proposition 2.2(2)); on monochrome I one takes $\mathcal{A}(I)$, resp. $\mathcal{C}(I)$.

Theorem 2.8 ([BDH])**.** *(A) If $\mathcal{A}, \mathcal{B}, \mathcal{C}$ are irreducible nets and $\mathcal{B}$ has finite index, then $D \circledast_{\mathcal{B}} E$ is a defect. (B) Under the same hypotheses $H_0(D \circledast_{\mathcal{B}} E) \cong H_0(D) \boxtimes_{\mathcal{B}(I)} H_0(E)$. (Weak identity) $\mathrm{Id}_{\mathcal{A}} \circledast_{\mathcal{A}} \mathrm{Id}_{\mathcal{A}}$ is the weak identity of Definition 2.6.*

2.5 Braided products of Q-systems

Definition 2.9 ([BKLR, Def. `d:Qprod`]) For Q-systems $\mathbf{A} = (\theta^A, w^A, x^A)$, $\mathbf{B} = (\theta^B, w^B, x^B)$ in a braided C* tensor category, the braided products $\mathbf{A} \times^{\pm} \mathbf{B}$ have object $\theta^A\theta^B$ and

$$w = w^A \times w^B, \qquad x^{\pm} = (1_{\theta^A} \times \varepsilon^{\pm}_{\theta^A,\theta^B} \times 1_{\theta^B}) \circ (x^A \times x^B).$$

Here $\varepsilon_{\rho,\sigma} = 1$ if ρ is localized to the right of σ and $\varepsilon^-_{\rho,\sigma} = \varepsilon^*_{\sigma,\rho}$. The corresponding extension is in general neither irreducible nor a factor, and contains the extensions of $\mathbf{A}$ and $\mathbf{B}$ as intermediate algebras. Its generator is $v^{\pm} = v^A v^B$ with $v^B v^A = \iota(\varepsilon^{\pm}_{\theta^A,\theta^B}) v^A v^B$ [BKLR, Lemma `l:braidext`].

Theorem 2.10 ([BKLR, Thm. `t:centchar`]) *Let $\mathbf{A}, \mathbf{B}$ be simple Q-systems in a modular tensor category and $Z[\mathbf{A}]$, $Z[\mathbf{B}]$ their full centres. For the extension $N \subset M$ of either $Z[\mathbf{A}] \times^{\pm} Z[\mathbf{B}]$, $M' \cap M = \iota(N)' \cap M = \mathrm{Hom}(\iota, \iota)$, and the minimal central projections $E_{\mathbf{m}}$ are in bijection with the irreducible $\mathbf{A}$–$\mathbf{B}$-bimodules $\mathbf{m}$.*

Proposition 2.11 ([BKLR, Prop. `p:fullsimple`]) *$Z[\mathbf{A}]$ is irreducible iff $\mathbf{A}$ is simple. More generally* $\dim \mathrm{Hom}(\mathrm{id} \otimes \mathrm{id}, Z[\mathbf{A}])$ *equals the dimension of the centre of the extension of $\mathbf{A}$.*

Definition 2.12 ([BKLR, Defs. `d:bimprod`, Prop. `p:bimprod`]) For an $\mathbf{A}$–$\mathbf{B}$-bimodule $\mathbf{m}_1$ and a $\mathbf{B}$–$\mathbf{C}$-bimodule $\mathbf{m}_2$ the bimodule tensor product $\mathbf{m}_1 \otimes_{\mathbf{B}} \mathbf{m}_2$ is the $\mathbf{A}$–$\mathbf{C}$-bimodule on the range of a projection in $\beta_1\beta_2$ built from m_1, m_2, w^B; it corresponds to composition of the homomorphisms $\varphi_1 \circ \varphi_2 \colon M^{\mathbf{C}} \to M^{\mathbf{A}}$.

2.6 Conventions and standing hypotheses

C1. *Orientation.* The left-to-right order of BDH's intervals is the increasing coordinate of the line, which is the future direction in BKLR's chiral convention ($\varepsilon_{\rho,\sigma} = 1$ if ρ is localized to the right of σ). Then the fused product is $Q \times^- Q'$ (Section 6.3); the opposite choice gives $Q \times^+ Q'$ and changes nothing else.

C2. *Reversed charts.* On a reversed chart a defect is evaluated by $D(-I) := D(I)^{\mathrm{op}}$ (BDH), and for a fusion $(D \circledast E)(-I) := (D \circledast E)(I)^{\mathrm{op}}$, realized on the fused space by complex conjugation $\overline{H \boxtimes_C K} \cong \bar{H} \boxtimes_{C^{\mathrm{op}}} \bar{K}$.

C3. *Vacuum normalization.* $H_0(\varphi)\Omega = \Omega$ (Section 2.1).

Assumption 2.13. Fix defects $D = D_Q$, $E = D_{Q'}$ with $Q = (\theta, w, x)$, $Q' = (\theta', w', x')$, and a genuinely bicolored I with the charts of Definition 2.7. Put $P_1 := (I^{++})^\sharp$, $M_1 := D(I^{++}) = N_1 v$, $N_1 := \mathcal{A}(P_1)$, and $P_2 := ({}^{++}I)^\sharp$, $M_2 := E({}^{++}I) = N_2 v'$, $N_2 := \mathcal{A}(P_2)$. Let $\tilde{J} = [\frac{1}{2}, \frac{3}{2}] \subset I_\bullet^{++}$ be the image of J and $\tilde{C} := \mathcal{A}(\tilde{J}) \subset N_1$, $C := \mathcal{A}(J) \subset N_2$, and

$$K := (I^+)^\sharp = P_1 \setminus \tilde{J}, \qquad L := ({}^+I)^\sharp = P_2 \setminus J.$$

Then $D(K) = \mathcal{A}(K)v$, $E(L) = \mathcal{A}(L)v'$, $M_1 \cap \tilde{C}' = D(K)$, $M_2 \cap C' = E(L)$, and $F := (D \circledast E)(I) = D(K) \vee E(L)$ acts on $H := H_1 \boxtimes_C H_2$, $H_i = L^2(M_i)$. On the line, K and L abut at the corner (the two buffers $[0, \frac{1}{2}]_\bullet$ and $[-\frac{1}{2}, 0]_\circ$ together form the inserted interval of the weak identity), and J is the vertical edge hanging off the junction. Standing hypotheses:

S1. $\mathcal{A}$ is a conformal net with vacuum Ω, Reeh–Schlieder, strong additivity, Haag duality; θ, θ' are normal localized endomorphisms, θ localized in an interval inside K (trivial on $\tilde{J}$), θ' localized inside L (trivial on J).

S2. N_1, N_2 are type III factors. Neither finite index of $\mathcal{A}$ nor irreducibility of Q, Q' is used in Sections 4–6.

Two consequences are used repeatedly. *State matching:* the anti-homomorphism $\sigma\colon C \to \tilde{C}$ induced by $x \mapsto \frac{3}{2} - x$ is $\sigma(c) = \Theta_r c^* \Theta_r$ with $\Theta_r = H_0(r)$, $\Theta_r \Omega = \Omega$, $\Theta_r^2 = 1$ by (F2), C3; hence $\omega \circ \sigma = \omega$ on C. *Commutation:* θ, θ' commute elementwise (each is trivial where the other is localized) and v, v' commute, being elements of the commuting algebras $D(K)$, $E(L)$ of F.

2.7 Standard facts from the literature (cited without proof)

L1 The expectation E of an extension $N \subset Nv$ given by a Q-system is faithful, and it is the minimal expectation when the extension is a factor (Longo; Kosaki; Hiai).

L2 The minimal index is multiplicative for tensor products and compositions (Kosaki–Longo; Longo).

L3 For a finite-index inclusion $B \subset R$ of properly infinite factors with a faithful normal conditional expectation there are a Q-system (θ_B, w_B, x_B) in $\mathrm{End}(B)$ and $v_R \in R$ with $R = Bv_R$ and $v_R b = \theta_B(b) v_R$ (Longo).

3 Background II: tensor categories

3.1 Setting

$\mathcal{C} = \mathrm{DHR}(A)$ is a modular unitary fusion category ([KLM], [BKLR]). Put $\mathcal{Z} := \mathcal{C} \boxtimes \mathcal{C}^{\mathrm{opp}}$, realised as a full braided subcategory of $\mathrm{DHR}(\mathcal{A})$ (Lemma 7.4); for modular $\mathcal{C}$ it is braided equivalent to the Drinfeld centre $Z(\mathcal{C})$ [Müger]. All Q-systems below live in $\mathcal{Z}$. For a simple Q-system $\mathbf{X}$ in $\mathcal{C}$ write $\mathcal{D}_{\mathbf{X}} := {}_{\mathbf{X}}\mathcal{C}_{\mathbf{X}}$ for its bimodule category, a fusion category Morita equivalent to $\mathcal{C}$, so $Z(\mathcal{D}_{\mathbf{X}}) \simeq Z(\mathcal{C}) \simeq \mathcal{Z}$. Let $F_{\mathbf{X}}\colon \mathcal{Z} \to \mathcal{D}_{\mathbf{X}}$ be the resulting central tensor functor (concretely, α-induction) and $I_{\mathbf{X}}$ its adjoint. For simple $\mathbf{X}, \mathbf{Y}$ the category $\mathcal{B} = {}_{\mathbf{X}}\mathcal{C}_{\mathbf{Y}}$ is an invertible $\mathcal{D}_{\mathbf{X}}$–$\mathcal{D}_{\mathbf{Y}}$-bimodule category; its internal Homs are $\underline{\mathrm{Hom}}(\mathbf{n}, \mathbf{n}') = \bar{\mathbf{n}} \otimes_{\mathbf{X}} \mathbf{n}' \in \mathcal{D}_{\mathbf{Y}}$ (right action) and $\mathbf{n}' \otimes_{\mathbf{Y}} \bar{\mathbf{n}} \in \mathcal{D}_{\mathbf{X}}$ (left action). For a Q-system P, η_P, m, $\varepsilon_P = w^* \circ (\cdot)$ and $x = m^*$ denote unit, product, counit and coproduct; $\varepsilon_P(\eta_P) = d_P$. An element $q \in \mathrm{Hom}(1, P)$ is *central* if $m(q \otimes 1) = m(1 \otimes q)$.

3.2 Quoted facts

(T1) $Z[\mathbf{X}] \cong I_{\mathbf{X}}(1)$ as algebras in $\mathcal{Z}$; $Z[\mathbf{X}]$ is commutative, and $\mathrm{Hom}(1, Z[\mathbf{X}]) = \mathbb{C}$ for simple $\mathbf{X}$ ([BKLR], Proposition 2.11; [Kong–Runkel]; [Davydov, *Centre of an algebra*, Thm. 6.2]).

(T2) The braided products $\times^{\pm}$ are associative, and the braiding gives algebra isomorphisms $Q_1 \times^{\mp} Q_2 \cong Q_2 \times^{\pm} Q_1$ [BKLR].

(T3) For a left $\mathcal{D}$-module category $\mathcal{M}$ the full centre $Z(\mathcal{M}) \in Z(\mathcal{D})$ represents $Z \mapsto \mathrm{Nat}_{\mathcal{D}}(Z * (-), \mathrm{Id}_{\mathcal{M}})$, the module natural transformations [Davydov, §6].

(T4) Under $\mathcal{Z} \simeq Z(\mathcal{D}_{\mathbf{B}})$, $Z[\mathbf{C}]$ is the full centre of the $\mathcal{D}_{\mathbf{B}}$-module category ${}_{\mathbf{B}}\mathcal{C}_{\mathbf{C}}$; and the identifications $Z(\mathcal{D}_{\mathbf{X}}) \simeq \mathcal{Z} \simeq Z(\mathcal{D}_{\mathbf{Y}})$ satisfy $F_{\mathbf{X}}(Z) \otimes (-) \cong (-) \otimes F_{\mathbf{Y}}(Z)$ as bimodule functors on ${}_{\mathbf{X}}\mathcal{C}_{\mathbf{Y}}$ (Morita invariance of the centre and of the full centre; [Davydov, Thm. 6.2], [ENO], [EGNO, Ch. 7–8]).

(T5) For $X \in \mathcal{D}$, the module functor $I(X) * (-)$ is the free $\mathcal{D}$-module functor on $X \otimes (-)$: $\mathrm{Nat}_{\mathcal{D}}(I(X) * (-), G) \cong \mathrm{Nat}(X \otimes (-), G)$ [EGNO, §7.12].

(T6) I is a two-sided adjoint of F (semisimple, spherical), and $I(1)$ is a Frobenius algebra with the lax product and the oplax coproduct.

(T7) $X \mapsto I(X)$ is a tensor equivalence $\mathcal{D} \simeq \mathcal{Z}_{I(1)}$ [Davydov–Nikshych, *Picard crossed module*, §4.2], [DMNO]; and for a commutative algebra L and an algebra B receiving a central map from L, B-modules in $\mathcal{Z}_L$ are B-modules in $\mathcal{Z}$ [Pareigis], [Kirillov–Ostrik].

The full centre of an algebra and the "projection formula" below are also in [Davydov]; the group-theoretical full centres used in Appendix A are [Davydov, Prop. 9.5].

3.3 Q-systems as algebras

Lemma 3.1 (relative commutant as an algebra)**.** *For a Q-system (θ, w, x) with $M = Nv$, the map $a \mapsto r_a := \iota(a)v$ is a linear bijection from $\mathrm{Hom}(\theta, \mathrm{id}) = \{a \in N : \ a\theta(n) = na \ \forall n\}$ onto $\iota(N)' \cap M$, and $r_a r_{a'} = r_{a \star a'}$ with $a \star a' := a\,\theta(a')\,x$. In particular $\iota(N)' \cap M$ is a finite-dimensional C^*-algebra if θ has finite dimension, and it is isomorphic to its opposite.*

Proof. $r_a \iota(n) = \iota(a\theta(n))v = \iota(na)v = \iota(n) r_a$ if $a \in (\theta, \mathrm{id})$; conversely $\iota(b)v \in \iota(N)'$ forces $b\theta(n) = nb$ by injectivity of $b \mapsto bv$ (Lemma 4.3). Then $r_a r_{a'} = \iota(a)v\iota(a')v = \iota(a\theta(a'))vv = \iota(a\theta(a')x)v$. A finite-dimensional von Neumann algebra is a sum of matrix algebras, and $M_n(\mathbb{C}) \cong M_n(\mathbb{C})^{\mathrm{op}}$. □

Lemma 3.2. *$M' \cap M = \{r_a : a \in \mathrm{Hom}(\theta, \mathrm{id}),\ a\,x = \theta(a)\,x\}$, which under the identification of Lemma 3.1 is $\{q \in \mathrm{Hom}_{\mathcal{Z}}(1, \Theta) : \ m(q \otimes 1) = m(1 \otimes q)\} = \mathrm{End}_{\Theta\text{-}\Theta}(\Theta)$. Moreover $\mathrm{End}_{\Theta\text{-}\Theta}(\Theta) = \mathrm{End}(\mathrm{Id}_{\mathcal{N}})$, the natural endomorphisms, as a $\mathcal{Z}$-module functor, of the identity of $\mathcal{N} := \mathrm{Mod}_{\Theta}(\mathcal{Z})$, and this has dimension equal to the number of indecomposable $\mathcal{Z}$-module-category summands of $\mathcal{N}$.*

Proof. Elements of $\iota(N)' \cap M$ are the r_a; r_a is central iff it commutes with v, and $r_a v = \iota(a)vv = \iota(ax)v$, $vr_a = \iota(\theta(a))vv = \iota(\theta(a)x)v$; injectivity of $b \mapsto bv$ gives the first claim. In categorical terms $ax = \theta(a)x$ says that the two maps $Q \to Q$ obtained from a and the coproduct x agree, i.e. q commutes with left and right multiplication. The functor $P \mapsto - \otimes_{\Theta} P$ is an equivalence ${}_{\Theta}\mathcal{Z}_{\Theta} \simeq \mathrm{Fun}_{\mathcal{Z}}(\mathcal{N}, \mathcal{N})$ taking Θ to Id, and $\mathrm{End}(\mathrm{Id}_{\mathcal{N}^k}) = \mathbb{C}$ for an indecomposable module category $\mathcal{N}^k$ (its dual category has simple unit), so $\dim \mathrm{End}(\mathrm{Id}_{\mathcal{N}})$ is the number of indecomposable summands. □

3.4 The projection formula

Lemma 3.3 (projection formula for algebras)**.** *Let $F \colon \mathcal{Z} = Z(\mathcal{D}) \to \mathcal{D}$ be the forgetful functor with right adjoint I (lax monoidal as the right adjoint of a tensor functor), and let X be an algebra in $\mathcal{Z}$. Then the projection isomorphism $I(1) \otimes X \to I(F(X))$ is an isomorphism of algebras from $I(1) \times^{\pm} X$ (product defined with the braiding of $\mathcal{Z}$) to $I(F(X))$ (product from the lax structure of I and the multiplication of $F(X)$), for either sign. Consequently $\mathrm{Hom}_{\mathcal{Z}}(1, I(1) \times^{\pm} X) \cong \mathrm{Hom}_{\mathcal{D}}(1, F(X))$ as algebras.*

Proof. The projection map $\pi\colon I(1)\otimes X \to I(F(X))$ is adjoint to $F(I(1)\otimes X) = FI(1)\otimes F(X) \xrightarrow{\varepsilon\otimes 1} F(X)$, $\varepsilon\colon FI(1)\to 1$ the counit; it is an isomorphism (projection formula for central functors). By adjunction, the two maps $I(1)\otimes X\otimes I(1)\otimes X \to I(F(X))$ obtained by (a) braiding and multiplying in $\mathcal{Z}$ and then applying π, and (b) applying $\pi\otimes\pi$ and then multiplying in $I(F(X))$, are equal iff their adjoints $FI(1)\otimes FX\otimes FI(1)\otimes FX \to FX$ are equal. The adjoint of (b) is $\varepsilon\otimes\mu_X\circ(1\otimes\varepsilon\otimes 1)$-type: apply ε to both copies of $FI(1)$ and multiply the two copies of FX. The adjoint of (a) is: apply $F(c)$ to move the second $I(1)$ past the first X, then multiply, then apply $\varepsilon\otimes 1$. Now $F(c_{X,I(1)}) = \widehat{X}_{FI(1)}$ is the half-braiding of X evaluated at $FI(1)\in\mathcal{D}$, which is natural in the $\mathcal{D}$-variable; applied to the morphism $\varepsilon\colon FI(1)\to 1$ and using $\widehat{X}_1 = \mathrm{id}$ this gives $(\varepsilon\otimes 1)\widehat{X}_{FI(1)} = 1\otimes\varepsilon$. So ε can be applied to the second copy of $FI(1)$ before braiding, and then the braiding is trivial; the remaining expression is the adjoint of (b). For the opposite sign use the inverse half-braiding $\widehat{X}^{-1}_{(\cdot)}$, which is also natural. Units match by the same naturality. The last statement is that the adjunction $F\dashv I$ is monoidal on $\mathrm{Hom}(1,\cdot)$. □

3.5 Two consequences of the quoted facts

Proposition 3.4 ((T8): induction followed by restriction)**.** *For simple* $\mathbf{X},\mathbf{Y}$, $\mathcal{B} = {}_{\mathbf{X}}\mathcal{C}_{\mathbf{Y}}$ *and* $V\in\mathcal{D}_{\mathbf{X}}$,

$$F_{\mathbf{Y}}I_{\mathbf{X}}(V) \;\cong\; \bigoplus_{\mathbf{n}\in\mathrm{Irr}\,\mathcal{B}} \underline{\mathrm{Hom}}_{\mathcal{B}}(\mathbf{n}, V\otimes\mathbf{n})=\bigoplus_{\mathbf{n}}\bar{\mathbf{n}}\otimes_{\mathbf{X}} V\otimes_{\mathbf{X}}\mathbf{n},$$

naturally in V*. In particular* $F_{\mathbf{Y}}(Z[\mathbf{X}])\cong\bigoplus_{\mathbf{n}}\underline{\mathrm{End}}(\mathbf{n})$*, and this isomorphism is the canonical map* φ *given by the action of the full centre on* $\mathcal{B}$*; it is an isomorphism of algebras.*

Proof. For $X\in\mathcal{D}_{\mathbf{Y}}$: $\mathrm{Hom}(F_{\mathbf{Y}}I_{\mathbf{X}}V, X) = \mathrm{Hom}_{\mathcal{Z}}(I_{\mathbf{X}}V, I_{\mathbf{Y}}X)$ by (T6). By (T4) the right $\mathcal{D}_{\mathbf{Y}}$-action identifies $\mathcal{D}_{\mathbf{Y}}$ with $\mathrm{Fun}_{\mathcal{D}_{\mathbf{X}}}(\mathcal{B},\mathcal{B})$ (invertibility of $\mathcal{B}$), and $I_{\mathbf{X}}V$, $I_{\mathbf{Y}}X$ act by the module functors $F_{\mathbf{X}}I_{\mathbf{X}}(V)\otimes(-)$ and $(-)\otimes X$ (the latter after applying $F_{\mathbf{Y}}I_{\mathbf{Y}}$ and the counit, i.e. $\mathrm{Hom}_{\mathcal{Z}}(Z, I_{\mathbf{Y}}X) = \mathrm{Hom}_{\mathcal{D}_{\mathbf{Y}}}(F_{\mathbf{Y}}Z, X)$ is $\mathrm{Nat}_{\mathcal{D}_{\mathbf{X}}}(Z*(-),(-)\otimes X)$). By (T5) the first is the free module functor on $V\otimes(-)$, so

$$\mathrm{Hom}(F_{\mathbf{Y}}I_{\mathbf{X}}V, X)\cong\mathrm{Nat}(V\otimes(-),(-)\otimes X)=\bigoplus_{\mathbf{n}}\mathrm{Hom}_{\mathcal{B}}(V\otimes\mathbf{n},\mathbf{n}\otimes X)=\bigoplus_{\mathbf{n}}\mathrm{Hom}_{\mathcal{D}_{\mathbf{Y}}}(\bar{\mathbf{n}}\otimes V\otimes\mathbf{n}, X),$$

naturally in X and V; Yoneda gives the isomorphism. For $V=1$: by (T1), (T3), (T4) the identity of $F_{\mathbf{Y}}Z[\mathbf{X}]$ corresponds to the universal action of the full centre on each $\mathbf{n}$, so the isomorphism is φ; φ is an algebra map (it is built from an action), hence an algebra isomorphism. □

Remark 3.5 (the product in (T8))*.* For an algebra V, $F_{\mathbf{Y}}I_{\mathbf{X}}(V)$ is an algebra (lax structure of $I_{\mathbf{X}}$), and so is the right side (composition of internal Homs on each summand, zero between summands). For $V=1$ the isomorphism is multiplicative by Proposition 3.4. For general V, multiplicativity is the statement that under (T5) the lax structure of $I_{\mathbf{X}}$ corresponds to composition of free module functors; this is a mate computation which I describe but have not drawn. It is used in Theorem 8.2 (hence for the algebra structure of relative commutants and for irreducibility), and nowhere else.

Proposition 3.6 (modules over $I(1)\times W$)**.** *Let* $F=F_{\mathbf{X}}$, $I=I_{\mathbf{X}}$, $\mathcal{D}=\mathcal{D}_{\mathbf{X}}$*, and* A *an algebra in* $\mathcal{D}$*. Then* $\mathrm{Mod}_{I(A)}(\mathcal{Z})\simeq\mathrm{Mod}_A(\mathcal{D})$*, compatibly with the left actions of* $\mathcal{Z}$ *(on the right side through* F*). In particular* $\mathrm{Mod}_{I(1)\times^{\pm}W}(\mathcal{Z})\simeq\mathrm{Mod}_{F(W)}(\mathcal{D})$ *for an algebra* W *in* $\mathcal{Z}$*.*

Proof. By (T7), $I(A)$ is an algebra in $\mathcal{Z}_{I(1)}$ and $\mathrm{Mod}_A(\mathcal{D}) \simeq \mathrm{Mod}_{I(A)}(\mathcal{Z}_{I(1)}) = \mathrm{Mod}_{I(A)}(\mathcal{Z})$. The $\mathcal{Z}$-actions match through the projection isomorphism $Z \otimes I(M) \cong I(F(Z) \otimes M)$, which is $I(A)$-linear because the right $I(A)$-action involves only the right factor and the lax structure of I, which commutes with the projection map by the naturality argument of Lemma 3.3. The last statement follows with $A = F(W)$ and Lemma 3.3. □

Since F is dominant, natural endomorphisms of the identity of $\mathrm{Mod}_A(\mathcal{D})$ that are compatible with the $\mathcal{Z}$-action through F are compatible with the whole $\mathcal{D}$-action: if $X \xrightarrow{\iota} F(Z) \xrightarrow{\pi} X$ with $\pi\iota = 1$, then $\eta_{X\otimes M} = (\pi \otimes 1)\eta_{F(Z)\otimes M}(\iota \otimes 1) = 1 \otimes \eta_M$. Combined with Lemma 3.2: *the centre of the extension of $I(A)$ has dimension equal to the number of indecomposable $\mathcal{D}$-module summands of* $\mathrm{Mod}_A(\mathcal{D})$.

Remark 3.7 (signs). Theorem 1.2 produces $\times^-$ in convention C1. Every statement below holds for both signs: Lemma 3.3 is proved for both, (T2) exchanges them, full centres are commutative for c and c^{-1}, and the remaining lemmas are proved for either sign. If the DHR braiding corresponds to the reverse of the braiding of $Z(\mathcal{D}_{\mathbf{X}})$, one runs the same arguments over $\mathcal{D}_{\mathbf{X}}^{\otimes\mathrm{op}}$, using $Z(\mathcal{D})^{\mathrm{rev}} \simeq Z(\mathcal{D}^{\otimes\mathrm{op}})$; the conclusions are unchanged. So no matching of sign conventions is needed.

4 Algebraic lemmas on a Q-system

Throughout this section (θ, w, x) is a Q-system with scalar d in a type III factor N, $M = Nv$ the extension, $E(nv) = d^{-1}nw$, and $\omega_M = \omega \circ E$.

Lemma 4.1. *Put $V := d^{-1}xw$ and $\varphi(n) := V^*\theta(n)V$. Then*
(a) $V^*V = 1$ and $Va = \theta^2(a)V$ for all $a \in N$;
(b) φ is unital, completely positive and normal, $\varphi(\theta(a)n\theta(b)) = a\varphi(n)b$, and $\varphi \circ \theta = \mathrm{id}$;
(c) $\theta \circ \varphi$ is a conditional expectation onto $\theta(N)$;
(d) $v^*bv = w^*x^*\theta(b)xv$, $E(v^*bv) = d\,\varphi(b)$, $v^*v = d$; hence $\|\Lambda(av)\|^2 = d\,\omega(\varphi(a^*a))$ and, for $\Phi := \Lambda(v)/\sqrt{d}$, $\langle \Phi, n\Phi\rangle = \omega(\varphi(n))$;
(e) φ is faithful.

Proof. (a) $V^*V = d^{-2}w^*x^*xw = d^{-2}\,d\,w^*w = 1$, and $Va = d^{-1}x\theta(a)w = d^{-1}\theta^2(a)xw = \theta^2(a)V$. (b) Complete positivity is clear. Using $V^*\theta^2(a) = aV^*$ and $\theta^2(b)V = Vb$ one finds $\varphi(\theta(a)n\theta(b)) = V^*\theta^2(a)\theta(n)\theta^2(b)V = aV^*\theta(n)Vb$; for $n = 1$ this gives $\varphi\theta(a) = a$. (c) follows from (b) by Tomiyama. (d) $v^*bv = w^*x^*v\,b\,v = w^*x^*\theta(b)vv = w^*x^*\theta(b)xv$, so $E(v^*bv) = d^{-1}w^*x^*\theta(b)xw = d\varphi(b)$; and $v^*v = w^*x^*xv = d\,w^*v = d$. (e) If $\varphi(n^*n) = 0$ then $E((nv)^*nv) = 0$, so $nv = 0$ by faithfulness of E (L1), and $n = 0$ since $a \mapsto av$ is injective. □

Lemma 4.2. *In $L^2(M)$: $v\Phi = x\Phi$ and $\Omega = \Lambda(1) = d^{-1/2}v^*\Phi$.*

Proof. $vv = xv$ and $v^*v = d$. □

Lemma 4.3. *(a) $vv^* = x^*v$ and $E(vv^*) = d^{-1}$. (b) For $m = nv \in M$ one has $n = d\,E(mv^*)$, hence $\|n\| \le d\|v\|\,\|m\|$.*

Proof. (a) $vv^* = v\,w^*x^*v = \theta(w^*)\theta(x^*)\,vv = \theta(w^*)\,\theta(x^*)x\,v = \theta(w^*)\,xx^*\,v = x^*v$, using $vv = xv$, $\theta(x^*)x = xx^*$ and $\theta(w^*)x = 1$; then $E(x^*v) = d^{-1}x^*w = d^{-1}(w^*x)^* = d^{-1}$. (b) $E(nvv^*) = E(nx^*v) = d^{-1}nx^*w = d^{-1}n$. □

Lemma 4.4 (product Q-system)**.** *Let θ' be trivial on $\mathcal{A}(K)$ and θ trivial on $\mathcal{A}(L)$, with v, v' and the primed and unprimed generators commuting. Put $v'' := vv'$, $\theta'' := \theta\theta'$, $w'' := ww'$, $x'' := xx'$, $d'' := dd'$, $V'' := VV'$. Then:*

(i) $v''n = \theta''(n)v''$ *for* $n \in \mathcal{A}(K) \vee \mathcal{A}(L)$ *(sums of products);*

(ii) $w''^*v'' = 1$, $v''v'' = x''v''$, $v''^* = w''^*x''^*v''$, $v''^*v'' = d''$;

(iii) $x''\theta''(n) = \theta''^2(n)x''$, $w''n = \theta''(n)w''$, $w''^*x'' = 1$, $x''x'' = \theta''(x'')x''$, $w''^*w'' = x''^*x'' = d''$;

(iv) $\varphi''(ab) = \varphi(a)\varphi'(b)$ *for* $a \in \mathcal{A}(K)$, $b \in \mathcal{A}(L)$;

(v) $F_0 := \operatorname{span}\{ab\,v'' : a \in \mathcal{A}(K), b \in \mathcal{A}(L)\}$ *is a $*$-algebra containing v, v', $\mathcal{A}(K) \odot \mathcal{A}(L)$, and* $F_0 = \operatorname{span}\{(av)(bv')\} = D(K) \odot E(L)$.

Proof. (i)–(iii) follow from the commutation hypotheses, e.g. $v''v'' = vvv'v' = xx'vv'$ and $w''^*v'' = w'^*(w^*v)v' = 1$. (iv) $\varphi''(ab) = V'^*V^*\theta(a)\theta'(b)VV' = V^*\theta(a)V \cdot V'^*\theta'(b)V'$. (v) $(abv'')(a'b'v'') = a\theta(a')x\,b\theta'(b')x'\,v''$ and $(abv'')^* = w''^*x''^*\theta''(a^*b^*)v'' \in F_0$; $ab = (abw''^*)v''$ since $w''^*v'' = 1$; $w'^*v'' = v$, $w^*v'' = v'$, and $(av)(bv') = ab\,vv'$ since $vb = \theta(b)v = bv$. □

These are the relations of $\mathbf{A} \times^{\pm} \mathbf{B}$ in Definition 2.9 with $\varepsilon^{\pm} = 1$ on the pair; the Frobenius and normalization relations for the product are those of [BKLR, Def. `d:Qprod`].

Lemma 4.5. *Let θ be localized in an interval inside K. Then $w, x, V \in \mathcal{A}(K)$, $\theta(\mathcal{A}(K)) \subset \mathcal{A}(K)$ and $\varphi(\mathcal{A}(K)) \subset \mathcal{A}(K)$.*

Proof. For $c \in \mathcal{A}(K')$, $\theta(c) = c$, so $wc = cw$ and $xc = x\theta(c) = \theta^2(c)x = cx$; thus $w, x \in \mathcal{A}(K')' = \mathcal{A}(K)$ by Haag duality, and $V \in \mathcal{A}(K)$. For $a \in \mathcal{A}(K)$, $\theta(a)c = \theta(ac) = \theta(ca) = c\theta(a)$, so $\theta(a) \in \mathcal{A}(K)$, hence $\varphi(a) \in \mathcal{A}(K)$. □

5 The fused space

Notation as in Assumption 2.13. Let $\omega_{M_i} := \omega \circ E_i$, realize H_i as the GNS space of (M_i, ω_{M_i}) with $\Omega_i := \Lambda(1)$, and put $\Phi_1 := \Lambda(v)/\sqrt{d}$, $\zeta := \Lambda(v')/\sqrt{d'}$, $\hat{\Omega} := \Omega_1 \boxtimes \Omega_2$, $\hat{\Phi} := \Phi_1 \boxtimes \zeta$. The right C-structure on H_1 is $\xi \cdot c = \sigma(c)\xi$.

Lemma 5.1. *For $a \in D(K)$ the vectors $a\Omega_1$, $a\Phi_1$ are $\tilde{C}$-bounded with $R(a\Omega_1) = aR(\Omega_1)$, and $R(\Omega_1)$, $R(\Phi_1)\colon L^2(C) \to H_1$ are isometries.*

Proof. $\|\sigma(c)\Omega_1\|^2 = \omega(\sigma(c)^*\sigma(c)) = \omega(cc^*) = \|\Omega_C \cdot c\|^2$ by state matching. For Φ_1, $\|\sigma(c)\Phi_1\|^2 = \omega(\varphi(\sigma(c)^*\sigma(c))) = \omega(\sigma(c)^*\sigma(c))$ since θ is trivial on $\tilde{J}$ (Lemma 4.1(b),(d)). For a commuting with $\tilde{C}$: $\sigma(c)a\xi = a\sigma(c)\xi$. □

Proposition 5.2 (state identity)**.** *For $a \in \mathcal{A}(K)$, $b \in \mathcal{A}(L)$,*

$$\langle \hat{\Phi}, (a \otimes b)\hat{\Phi} \rangle = \langle \hat{\Omega}, (\varphi(a) \otimes \varphi'(b))\hat{\Omega} \rangle.$$

Consequently, for $f = \sum a_i b_i v''$, $f' = \sum a'_j b'_j v'' \in F_0$ one has $\langle f\hat{\Phi}, f'\hat{\Phi} \rangle = \langle \hat{\Phi}, f^*f'\hat{\Phi} \rangle$ with $f^*f' = \sum a''b''v''$ and $\langle \hat{\Phi}, (a''b''v'')\hat{\Phi} \rangle = \langle \hat{\Omega}, (\varphi(a''x) \otimes \varphi'(b''x'))\hat{\Omega} \rangle$.

Proof. Let $Y_a := R(\Phi_1)^* a R(\Phi_1) \in \mathrm{End}_C(L^2C_C) = C$. For $c_1, c_2 \in C$ with $\tilde{c}_i = \sigma(c_i)$ and $\theta(\tilde{c}_i) = \tilde{c}_i$,

$$\langle \Omega_C c_1, Y_a \Omega_C c_2 \rangle = \langle \tilde{c}_1\Phi_1, a\tilde{c}_2\Phi_1 \rangle = \omega(\varphi(\tilde{c}_1^* a \tilde{c}_2)) = \omega(\tilde{c}_1^*\varphi(a)\tilde{c}_2) = \langle \tilde{c}_1\Omega_1, \varphi(a)\tilde{c}_2\Omega_1 \rangle,$$

by Lemma 4.1(b),(d). The right side is the form defining $Y_{\varphi(a)} := R(\Omega_1)^*\varphi(a)R(\Omega_1)$ ($\varphi(a) \in \mathcal{A}(K)$ commutes with $\tilde{C}$ by Lemma 4.5), so $Y_a = Y_{\varphi(a)} \in C$. Since $b \in E(L)$ commutes with C, and $Y_a \in C$ commutes with v',

$$\langle \hat{\Phi}, (a \otimes b)\hat{\Phi} \rangle = \langle \zeta, Y_a b \zeta \rangle = d'^{-1}\omega\big(E'(v'^* b v' Y_a)\big) = \omega(\varphi'(b)Y_a),$$

and the same computation for $(\Omega_2, \varphi(a), \varphi'(b))$ gives the right-hand side. For the consequence, $v''\hat{\Phi} = (x \otimes x')\hat{\Phi}$ by Lemma 4.2, and products in F_0 are given by Lemma 4.4(v). □

Lemma 5.3 (Reeh–Schlieder for the extension strip)**.** *$D(K)\Omega_1 = \Lambda(\mathcal{A}(K)v)$ is dense in H_1, and $E(L)\Omega_2$ is dense in H_2.*

Proof. By Lemma 4.1(d), $\Lambda(av) \mapsto \sqrt{d}\,\theta(a)V\Omega$ is an isometry from $\Lambda(N_1 v) = \Lambda(M_1)$ onto $\overline{\{\theta(n)V\Omega : n \in N_1\}}$, so it suffices to show $\theta(n)V\Omega \in \overline{\{\theta(a)V\Omega : a \in \mathcal{A}(K)\}}$ for $n \in N_1$.

Step 1. Let $a \in \mathcal{A}(K)$, $\tilde{c} \in \mathcal{A}(\tilde{J})$. Since θ is trivial on $\tilde{J}$ and $V \in \mathcal{A}(K)$ commutes with $\tilde{c}$, $\theta(a\tilde{c})V\Omega = \theta(a)V\tilde{c}\Omega$. By Reeh–Schlieder choose $a_k \in \mathcal{A}(K)$ with $a_k\Omega \to \tilde{c}\Omega$. Using $Va_k = \theta^2(a_k)V$,

$$\theta(a)V\tilde{c}\Omega = \lim_k \theta(a)Va_k\Omega = \lim_k \theta(a)\theta^2(a_k)V\Omega = \lim_k \theta(a\theta(a_k))V\Omega,$$

and $a\theta(a_k) \in \mathcal{A}(K)$ by Lemma 4.5.

Step 2. $\operatorname{span}\mathcal{A}(K)\mathcal{A}(\tilde{J})$ is a unital $*$-algebra, strongly dense in $\mathcal{A}(K) \vee \mathcal{A}(\tilde{J}) = \mathcal{A}(P_1)$ by strong additivity. By Kaplansky choose bounded $n_\alpha \to n$ strongly in this span; θ is normal, so $\theta(n_\alpha)V\Omega \to \theta(n)V\Omega$, and each $\theta(n_\alpha)V\Omega$ lies in the closure by Step 1. The E-side is identical. □

Lemma 5.4 (fusion norm against the commutant of the middle algebra)**.** *If $\zeta_1 \in H_1$ is $\tilde{C}$-bounded and $b \in M_2 \cap C'$, then $\|\zeta_1 \boxtimes b\Omega_2\| \le \|b\|\,\|\zeta_1\|$.*

Proof. Let $X := R(\zeta_1)^*R(\zeta_1) \in C$. Since b commutes with X and $X^{1/2}$,

$$\|\zeta_1 \boxtimes b\Omega_2\|^2 = \langle b\Omega_2, Xb\Omega_2 \rangle = \|bX^{1/2}\Omega_2\|^2 \le \|b\|^2\omega(X) = \|b\|^2\|R(\zeta_1)\Omega_C\|^2 = \|b\|^2\|\zeta_1\|^2,$$

using state matching for $\langle \Omega_2, X\Omega_2 \rangle = \langle \Omega_C, X\Omega_C \rangle$. □

The bound controls the fusion norm by the Hilbert norm of ζ_1, although elements of $D(K)$ approximating a vector $\tilde{c}\Phi_1$ in Hilbert norm cannot be norm-bounded (otherwise $\tilde{c} \in D(K) \cap \mathcal{A}(\tilde{J}) = \mathbb{C}$).

Theorem 5.5 (cyclicity)**.** *Under Assumption 2.13, $\hat{\Omega}$ is cyclic for $\overline{\operatorname{span}}\{(a \otimes 1)(1 \otimes b) : a \in D(K), b \in E(L)\}$, hence for F, and $\hat{\Phi}$ is cyclic for F_0 and for F: $\overline{F_0\hat{\Phi}} = \overline{F\hat{\Phi}} = H$.*

Proof. Vectors $\xi \boxtimes \eta$ with ξ bounded span a dense subspace of H. (i) By Lemma 5.3 choose $b_n \in E(L)$ with $b_n\Omega_2 \to \eta$; then $\|\xi \boxtimes (\eta - b_n\Omega_2)\| \le \|R(\xi)\|\|\eta - b_n\Omega_2\| \to 0$. (ii) Fix $b \in E(L)$ (it commutes with C). Choose $a_m \in D(K)$ with $a_m\Omega_1 \to \xi$; $\xi - a_m\Omega_1$ is $\tilde{C}$-bounded (Lemma 5.1) and Lemma 5.4 gives $\|(\xi - a_m\Omega_1) \boxtimes b\Omega_2\| \le \|b\|\|\xi - a_m\Omega_1\| \to 0$. (iii) $a\Omega_1 \boxtimes b\Omega_2 = (a \otimes 1)(1 \otimes b)\hat{\Omega}$. (iv) By Lemma 4.2, $\hat{\Omega} = (dd')^{-1/2}(v^* \otimes v'^*)\hat{\Phi}$ with $v^* \otimes 1, 1 \otimes v'^* \in F_0$, so $\hat{\Omega} \in F_0\hat{\Phi}$. □

The argument uses vacuum entanglement across J: in a finite-dimensional tracial toy with a product vacuum the fused space is larger by a factor $\dim C$.

Corollary 5.6. *$H = \overline{F_0\hat{\Phi}}$, and $f\hat{\Phi} \mapsto f\Lambda''(v'')/\sqrt{d''}$ is an isometry with dense range from $F_0\hat{\Phi}$ into the GNS space of $F_0^{\mathrm{alg}} = \{\sum abv''\}$ with the state $\omega_{\circledast} \circ E''$, $E''(abv'') = d''^{-1}abw''$, where $\omega_{\circledast} = \langle\hat{\Omega}, \cdot\,\hat{\Omega}\rangle$ on $N_{\circledast} := \mathcal{A}(K) \vee \mathcal{A}(L)$. Thus H is $L^2(M'')$ for $M'' = N_{\circledast}v''$ once θ'' is known to be normal on $N_{\circledast}$ (Section 6).*

Proof. Proposition 5.2 gives the inner products of $f\hat{\Phi}$ as $\omega_{\circledast}(\varphi(a''x)\varphi'(b''x')) = \omega_{\circledast}(\varphi''(a''b''x''))$ (Lemma 4.4(iv)), which is the state of $\Lambda''(v'')/\sqrt{d''}$ on F_0^{alg} by Lemma 4.1(d) for the product Q-system. Density is Theorem 5.5. □

6 Identification with the braided product

6.1 Vacuum gluing

Let $\Theta_r = H_0(r)$ for the reflection $r\colon x \mapsto \frac{3}{2} - x$, and J_R the modular conjugation of $(\mathcal{A}(R), \Omega)$. For the identity defects $H_1 = L^2(N_1) = H_0$, $\Omega_1 = \Omega$, and $\sigma(c) = \Theta_r c^* \Theta_r$.

Proposition 6.1 (vacuum gluing)**.** *Let $h := r_J \circ r$ (an orientation-preserving Möbius map; $J = [0,1]$ in the coordinate of the second factor) and $\alpha_h := \operatorname{Ad} U(h)$. For $z \in \mathcal{A}(K)$ the C-valued inner product $Y_z := R(\Omega_1)^* z R(\Omega_1)$ equals $\alpha_h(z) \in \mathcal{A}(h(K)) \subset \mathcal{A}(J)$, where $h(K)$ is adjacent to the right end of J, hence to L. For $a, a' \in \mathcal{A}(K)$, $b, b' \in \mathcal{A}(L)$,*

$$\langle a\Omega_1 \boxtimes b\Omega_2, a'\Omega_1 \boxtimes b'\Omega_2\rangle = \langle \alpha_h(a)b\,\Omega, \alpha_h(a')b'\,\Omega\rangle_{H_0},$$

so $W\colon a\Omega_1 \boxtimes b\Omega_2 \mapsto \alpha_h(a)b\Omega$ extends to a unitary $L^2(N_1) \boxtimes_{\tilde{C}} L^2(N_2) \to H_0$, and $\operatorname{Ad} W$ is an isomorphism $N_{\circledast} = \mathcal{A}(K) \vee \mathcal{A}(L) \to \mathcal{A}(h(K)) \vee \mathcal{A}(L) = \mathcal{A}(h(K) \cup L)$ with $\operatorname{Ad}W(a \otimes 1) = \alpha_h(a)$ and $\operatorname{Ad}W(1 \otimes b) = b$.

Proof. Step 1: $R(\Omega_1)$ is unitary. Take $L^2(C) = H_0$, $\Omega_C = \Omega$ (Ω is cyclic and separating for $\mathcal{A}(J)$), so $\Omega_C \cdot c = J_J c^* \Omega$ with $J_J = U(r_J)$ by (F1). Then $R(\Omega_1)(J_J c^*\Omega) = \sigma(c)\Omega = \Theta_r c^* \Omega$, so $R(\Omega_1) = \Theta_r J_J = U(r \circ r_J)$ on a dense set; it is onto since $\mathcal{A}(\tilde{J})\Omega$ is dense. *Step 2.* $Y_z = J_J \Theta_r z \Theta_r J_J$; the composite of the two antilinear conjugations is linear and equals $\operatorname{Ad} U(r_J \circ r)$, so $Y_z = \alpha_h(z)$. Since $r_J(x) = x/(2x-1)$ maps $[1,\infty)$ onto $(\frac{1}{2}, 1]$, $h(K) \subset J$ and h sends the corner $\frac{1}{2}$ of K to the right end of J. *Step 3.* $\langle a\Omega_1 \boxtimes b\Omega_2, a'\Omega_1 \boxtimes b'\Omega_2\rangle = \langle b\Omega_2, Y_{a^*a'} b'\Omega_2\rangle = \langle \Omega, b^* \alpha_h(a^*a') b'\Omega\rangle$, using that $Y \in \mathcal{A}(J)$ commutes with $b \in \mathcal{A}(L)$ and that $\mathcal{A}(h(K))$ commutes with $\mathcal{A}(L)$. By Theorem 5.5 with $\theta = \mathrm{id}$, $\hat{\Omega}$ is cyclic for $\mathcal{A}(K) \odot \mathcal{A}(L)$, and Ω is cyclic for $\mathcal{A}(h(K)) \odot \mathcal{A}(L)$ by Reeh–Schlieder; so W is a unitary intertwining the algebraic actions, and strong additivity gives $\mathcal{A}(h(K)) \vee \mathcal{A}(L) = \mathcal{A}(h(K) \cup L)$. □

Since K and L do not lie in a common coordinate, the weak identity of [BDH] holds here in the twisted sense of h; as $\alpha_h\colon \mathcal{A}(K) \to \mathcal{A}(h(K))$ is a net isomorphism preserving order, this is the weak identity up to relabeling.

Lemma 6.2. *$N_{\circledast} \subset \mathbf{B}(H)$ is isomorphic to $(\mathrm{Id} \circledast \mathrm{Id})(\tilde{I})$, via restriction to the invariant subspace $H_{\circledast} = L^2(N_1) \boxtimes_{\tilde{C}} L^2(N_2) \ni \hat{\Omega}$, and $N_{\circledast} \cong \mathcal{A}(h(K) \cup L)$ by Proposition 6.1.*

Proof. $L^2(N_1) \subset L^2(M_1)$ is invariant under $\mathcal{A}(K)$ and the right C-structure, so $H_{\circledast} \subset H$ is invariant under $\mathcal{A}(K) \odot \mathcal{A}(L)$ and the algebra generated there is $(\mathrm{Id} \circledast \mathrm{Id})(\tilde{I})$. The vector $\hat{\Omega} \in H_{\circledast}$ is cyclic for $N_{\circledast}$ (Theorem 5.5, $\theta = \mathrm{id}$) and $\langle\hat{\Omega}, \cdot\,\hat{\Omega}\rangle$ is the transported vacuum state of $\mathcal{A}(h(K) \cup L)$, which is faithful; hence restriction to $H_{\circledast}$ is faithful. □

Lemma 6.3 (transport of a Q-system by a Möbius map)**.** *Let (θ, w, x) be a Q-system with θ localized in an interval Θ, h an orientation-preserving Möbius map and $\hat{\Theta} \supset \Theta$ slightly larger. Choose an interval I' containing $\hat{\Theta} \cup h\hat{\Theta}$ in its interior. There is a unitary $u \in \mathcal{A}(I')$ with $\mathrm{Ad}(u) = \alpha_h$ on $\mathcal{A}(\hat{\Theta})$, and with $s := u\theta(u)^*$,*

$$\theta^h := \alpha_h \theta \alpha_h^{-1} = \mathrm{Ad}(s)\theta, \qquad \alpha_h(w) = sw, \qquad \alpha_h(x) = s\theta(s)xs^*.$$

Hence $Q^h = (\theta^h, \alpha_h(w), \alpha_h(x))$ is equivalent to Q (Definition 2.4), and $Q^h \times^{\pm} Q' \cong Q \times^{\pm} Q'$ by naturality of the braiding.

Proof. $h|_{\hat{\Theta}}$ is an orientation-preserving embedding into the interior of I', so it extends to $\varphi \in \mathrm{Diff}_+(I')$ equal to the identity near $\partial I'$; $\mathcal{A}(\varphi) = \alpha_h$ on $\mathcal{A}(\hat{\Theta})$ and $\mathcal{A}(\varphi) = \mathrm{Ad}(u)$ with $u \in \mathcal{A}(I')$ by inner covariance. As θ is localized in $\Theta \subset I'$, $\theta(u) \in \mathcal{A}(I')$. For $a \in \mathcal{A}(\hat{\Theta})$, $\theta^h(a) = u\theta(u^*au)u^* = s\theta(a)s^*$. From $wa = \theta(a)w$ one gets $\theta(u)^*w = wu^*$, so $sw = uwu^* = \alpha_h(w)$. From $x\theta(a) = \theta^2(a)x$ with $a = u$, $\theta^2(u)^*x\theta(u) = x$, so $s\theta(s)xs^* = u\theta^2(u)^*x\theta(u)u^* = uxu^* = \alpha_h(x)$. These are the formulas of Definition 2.4. □

6.2 The net-level theorem

Theorem 6.4 (= Theorem 1.2, net level)**.** *Under Assumption 2.13 put $K^h := h(K)$ and $Q''_h := Q^h \times Q' = (\theta^h\theta', \alpha_h(w)w', \alpha_h(x)x')$, a Q-system of an interval $[0,1]''$ containing the supports of θ^h (to the left) and θ'. Then for every genuinely bicolored I there is a $*$-isomorphism*

$$(D_Q \circledast D_{Q'})(I) \cong D_{Q''_h}(I) = \big\langle \mathcal{A}(I^h_\circ \cup [0,1]'' \cup I_\bullet),\, v'' \big\rangle, \qquad I^h_\circ := h(I_\circ),$$

carrying $\mathcal{A}(K) \to \mathcal{A}(K^h)$ by α_h, $\mathcal{A}(L) \to \mathcal{A}(L)$, $v \mapsto w'^*v''$, $v' \mapsto \alpha_h(w)^*v''$, compatible with shrinking the outer parts of I; and $H_0(D_Q) \boxtimes_{\mathcal{A}(J)} H_0(D_{Q'}) \cong L^2(D_{Q''_h}(I))$ intertwining the actions.

Proof. By Proposition 6.1 and Lemma 6.2, $N_\circledast \cong \mathcal{A}(K^h \cup L) = \mathcal{A}(I^h_\circ \cup [0,1]'' \cup I_\bullet)$, carrying θ, θ' to θ^h, θ' (determined by their values on the generating subalgebras $\mathcal{A}(K), \mathcal{A}(L)$, where the isomorphism is α_h and the identity). Since θ^h and θ' are localized normal endomorphisms on adjacent intervals, $\theta'' = \theta^h\theta'$ is normal on $N_\circledast$, so E'' exists (this settles the normality needed in Corollary 5.6). Lemma 4.4 gives the Q-system relations with $v'' \in F$, hence $F_0 \cong N_\circledast v''$, which is the algebra of $D_{Q''_h}(I)$ (injectivity of $a \mapsto av''$ follows from E''). Corollary 5.6 gives the unitary $H \to L^2(M'')$ intertwining F_0 with $N_\circledast v''$, and weak closures give $F \cong M''$. For sub-intervals, $F(I_1) = D(I_1^+) \vee E({}^+I_1)$ is generated by the correspondingly smaller strips and v, v', and h does not depend on I_1. □

6.3 Sign of the braided product

In the fused line θ sits to the left of θ'. Then $\varepsilon_{\theta',\theta} = 1$ and $\varepsilon^-_{\theta,\theta'} = \varepsilon^*_{\theta',\theta} = 1$, so $x'' = xx'$ is x^- of Definition 2.9: Q''_h is $Q^h \times^- Q'$ under C1. With the opposite orientation convention one obtains $\times^+$.

6.4 Change of chart; the fused object is a defect

Lemma 6.5 (change of chart)**.** *There is a Q-system $\hat{Q} = (\hat{\theta}, \hat{w}, \hat{x})$ in $\mathcal{A}([0,1])$ and a natural isomorphism $D_Q \circledast D_{Q'} \cong D_{\hat{Q}}$ on genuinely bicolored intervals with orientation-preserving, colour- and coordinate-preserving embeddings.*

Proof. Write $\Lambda_I := h(I_\circ) \cup [0,1]'' \cup I_\bullet$, the interval in which $D_{Q''_h}(I) = \langle \mathcal{A}(\Lambda_I), v'' \rangle$ lives, and recall $I^\sharp = I_\circ \cup [0,1] \cup I_\bullet$, glued by the local coordinate (white side $x \mapsto x$, black side $x \mapsto x+1$). Choose an orientation-preserving diffeomorphism $\kappa\colon [0,1] \to [0,1]''$ equal to h near 0 and to the translation matching the black chart near 1 (prescribe the two jets and interpolate a positive derivative). Define $g_I\colon I^\sharp \to \Lambda_I$ by h on $I_\circ$, κ on $[0,1]$, the identity on $I_\bullet$; the three pieces match to infinite order, so g_I is a diffeomorphism. Put $\hat\theta := \alpha_\kappa^{-1}\theta''\alpha_\kappa$, $\hat w := \alpha_\kappa^{-1}(w'')$, $\hat x := \alpha_\kappa^{-1}(x'')$, with $\alpha_\kappa = \mathcal{A}(\kappa)$; $\hat Q$ is a Q-system in $\mathcal{A}([0,1])$ because θ'', w'', x'' lie in $\mathcal{A}([0,1]'')$ (Lemma 4.5). Then $\mathcal{A}(g_I)$ restricts to α_κ on $\mathcal{A}([0,1])$, so $a \mapsto \mathcal{A}(g_I)(a)$, $\hat v \mapsto v''$ is an isomorphism $\chi_I\colon D_{\hat Q}(I) \to D_{Q''_h}(I)$.

Naturality. For $f\colon J \to I$ preserving orientation, colours and the local coordinate, f is the identity near the colour point; let $f^\sharp\colon J^\sharp \to I^\sharp$ be f on the outer parts and the identity on the strip, and $\hat f\colon \Lambda_J \to \Lambda_I$ be hfh^{-1} on $h(J_\circ)$, the identity on $[0,1]''$ and f on $J_\bullet$. On each piece $g_I f^\sharp = \hat f g_J$, so χ commutes with the structure maps. The isomorphism of Theorem 6.4 is given on generators independently of I (α_h on $\mathcal{A}(K)$, the identity on $\mathcal{A}(L)$), and the structure map of the fusion for f acts on $\mathcal{A}(K_J)$ by $\mathcal{A}(f_K)$, so $\Psi_I(D_Q \circledast D_{Q'})(f)(a) = \alpha_h\mathcal{A}(f_K)(a) = \mathcal{A}(hf_Kh^{-1})\alpha_h(a) = D_{Q''_h}(\hat f)\Psi_J(a)$, and likewise on $\mathcal{A}(L)$ and on v. □

Proposition 6.6. *Under Assumption 2.13, $D_Q \circledast D_{Q'}$ is naturally isomorphic, as a functor on coordinate-bicolored intervals, to the defect $D_{\hat Q}$; in particular it satisfies all defect axioms. No finite-index hypothesis on $\mathcal{A}$ is used.*

Proof. Orientation-reversing morphisms enter only through reversed coordinates via $D(-I) = D(I)^{\mathrm{op}}$, and the same formula defines $(D \circledast E)(I)$ for every local coordinate; by C2 the structure maps on orientation-reversing morphisms are conventions, not computations in the vacuum sector. For D_Q, v^* plays the role of v in $D_Q(I)^{\mathrm{op}}$: from $va = \theta(a)v$, $av^* = v^*\theta(a)$. The isomorphisms Ψ_I of Theorem 6.4 are given on generators independently of I, hence are natural for orientation-preserving embeddings (Lemma 6.5); setting Ψ_{-I} to be the op-transport of Ψ_I gives a natural isomorphism on all coordinate-bicolored intervals. A functor isomorphic to a defect is a defect, and $D_{\hat Q}$ is a defect by Proposition 2.5. □

Remark 6.7 (BDH). prove that $D \circledast_{\mathcal{B}} E$ is a defect when $\mathcal{B}$ has finite index and conjecture it in general; Proposition 6.6 verifies it for Q-system defects over any conformal net, modulo convention C2.

7 Consequences of Theorem 1.2

7.1 Irreducible and reducible products

By Theorem 6.4, $(D_Q \circledast D_{Q'})(I) \cong N''v''$ even if Q''_h is reducible: irreducibility of Q, Q' is not used in Sections 4–6. By [BKLR] the centre of M'' is $\{\iota(q)v'' : q \in \mathrm{Hom}(\theta'', \mathrm{id})$ satisfying the centre condition$\}$, a finite-dimensional space. Each such q intertwines θ'' with id, hence commutes with $\mathcal{A}$ of anything away from the support of θ'', and by Haag duality lies in $\mathcal{A}$ of an interval around that support. The minimal central projections z_k therefore do not depend on the outer parts $I_\circ, I_\bullet$ and are compatible with the embeddings $D(J) \to D(I)$ (for embeddings which move the outer parts, $D(f)$ is inner by Proposition 2.3 and acts trivially on the centre).

Lemma 7.1. *Let D be an $\mathcal{A}$–$\mathcal{A}$ defect and $z_I \in Z(D(I))$ central projections, one for each genuinely bicolored I of orientation type (white, black), with* (N)*: $D(f)(z_J) = z_I$ for every*

orientation-preserving embedding $f\colon J \to I$ of such intervals. Then $\{z_I\}$ extends uniquely to a natural family on all genuinely bicolored intervals, and $I \mapsto z_I D(I)$ (with $\mathcal{A}$, $\mathcal{B}$ on monochrome intervals) is a defect zD with $D = zD \oplus (1-z)D$.

Proof. Extension. Realize every (white, black) interval as a subinterval K of a bicolored circle S; two realizations differ by an orientation-preserving embedding, under which $\{z_I\}$ is natural by (N). For the (black, white) interval jK put $z_{jK} := D(j|_K)(z_K)$, a central projection since $D(j|_K)$ is an anti-isomorphism. Functoriality gives naturality under inclusions, and a general orientation-reversing embedding is j followed or preceded by orientation-preserving ones. *Functor.* Put $(zD)(f)(a) := z_I D(f)(a)$ on $z_J D(J)$; this is a homomorphism of corner algebras by naturality, injective since $D(f)$ is injective and $D(f)(z_J) = z_I$. For monochrome J set $(zD)(f)(a) := z_I D(f)(a)$ on $\mathcal{A}(J)$; injective because $\mathcal{A}(J)$ is a factor and $z_I \neq 0$. Isotony and locality are inherited, and strong additivity follows from $z_I(ab) = (z_I a)(z_I b)$. *Vacuum sector.* $L^2((zD)(I)) = P_z L^2 D(I)$ with P_z the left (= right, z_I being central) multiplication by z_I; α, β commute with P_z. For monochrome J, $\alpha \otimes \beta$ extends to a normal representation π of $D(J \cup \bar{J})$ by the axiom for D, with $\pi(D(J \cup \bar{J})) \subset \{\alpha(D(J)), \beta(D(\bar{J}))\}''$ by strong additivity, which commutes with P_z; so $\pi(\cdot)P_z$ is the extension required for zD. □

Proposition 7.2. *$D_Q \circledast D_{Q'} \cong \bigoplus_k z_k D_{Q''_h}$, a finite direct sum of irreducible defects, one for each minimal central projection z_k of M''.*

Proof. The z_k satisfy (N) by the preceding discussion, so Lemma 7.1 applies to each z_k and to $1 - z_k$; each $z_k D_{Q''_h}(I)$ is a factor because z_k is minimal. This agrees with [BDH, composite of semisimple defects is semisimple], which gives finiteness independently. □

The identification of the summands $z_k D$ with the reduced Q-systems of [BKLR] is not needed for the defect statement.

7.2 Full centres and phase boundaries

Let A be a completely rational conformal net with $\mathcal{C} = \mathrm{DHR}(A)$ modular (non-degeneracy of the DHR braiding for completely rational nets: [BKLR], [KLM, Cor. 32]), $\bar{A}(I) := A(rI)$ the conjugate net, and $\mathcal{A} := A \otimes \bar{A}$.

Lemma 7.3. *Let A, B be completely rational conformal nets on Hilbert spaces H_A, H_B. Then $A \otimes B$ is completely rational (in the sense of [KLM]: Haag duality, strong additivity, split property, finite μ-index).*

Proof. Haag duality: $(A(I)\bar{\otimes}B(I))' = A(I')\bar{\otimes}B(I')$ by Tomita's commutation theorem. Strong additivity: for adjacent I_1, I_2 with $I = I_1 \cup I_2$ the algebra generated by $A(I_i)\bar{\otimes}B(I_i)$ contains $a \otimes 1$, $1 \otimes b$ for $a \in A(I_1) \cup A(I_2)$, $b \in B(I_1) \cup B(I_2)$, hence equals $A(I)\bar{\otimes}B(I)$. Split: for disjoint I_1, I_2, $(A \otimes B)(I_1) \vee (A \otimes B)(I_2) = (A(I_1) \vee A(I_2))\bar{\otimes}(B(I_1) \vee B(I_2)) \cong (A \otimes B)(I_1)\bar{\otimes}(A \otimes B)(I_2)$. Index: for $E = I_1 \cup I_2$ one has $(A \otimes B)(E) = A(E)\bar{\otimes}B(E)$ and likewise for the dual algebras, so $\mu_{A\otimes B} = \mu_A \mu_B < \infty$ by multiplicativity of the minimal index (L2). Reeh–Schlieder: $\overline{A(I)\Omega_A} \otimes \overline{B(I)\Omega_B} = H_A \otimes H_B$. For $B = \bar{A}$ the same applies, $\bar{A}$ being a relabeling of A. □

Lemma 7.4. *(i) $\bar{A}$ is a conformal net in the sense of [BDH] with $\mathrm{DHR}(\bar{A}) \simeq \mathcal{C}^{\mathrm{opp}}$. (ii) The functor $\mathcal{C} \boxtimes \mathcal{C}^{\mathrm{opp}} \to \mathrm{DHR}(A \otimes \bar{A})$, $\rho \boxtimes \bar{\sigma} \mapsto \rho \otimes \bar{\sigma}$ is a full and faithful braided tensor functor.*

Proof. (i) Reversing orientation is an involutive functor on intervals, the identity on underlying embeddings and on orientation-preserving morphisms; put $\bar{A} := A \circ$ (orientation reversal), so $\bar{A}(I) \cong A(I)^{\mathrm{op}}$. Isotony, locality, strong additivity, the split property and inner covariance concern subintervals and orientation-preserving maps, hence are unchanged, and the vacuum axiom holds with $H_0(S, \bar{A}) := H_0(\bar{S}, A)$. In $\bar{A}$, "ρ is to the right of σ" means "to the left" in A, so by Definition 2.9 $\varepsilon^{\bar{A}}_{\rho,\sigma} = \varepsilon^{A}_{\sigma,\rho}{}^{*} = \varepsilon^{A,-}_{\rho,\sigma}$, i.e. $\mathrm{DHR}(\bar{A}) = \mathcal{C}^{\mathrm{rev}} \simeq \mathcal{C}^{\mathrm{opp}}$. (ii) For localized ρ_i, σ_i in the two legs, with $N_1 = (\rho_1 \oplus \sigma_1)(M_1)$, $N_2 = (\rho_2 \oplus \sigma_2)(M_2)$, one has $(N_1 \bar{\otimes} N_2)' = N_1' \bar{\otimes} N_2'$, and cutting by the projections onto the summands gives $\mathrm{Hom}(\rho_1 \otimes \rho_2, \sigma_1 \otimes \sigma_2) = \mathrm{Hom}(\rho_1, \sigma_1) \otimes \mathrm{Hom}(\rho_2, \sigma_2)$. The functor is braided since $\rho \otimes 1$ and $1 \otimes \bar{\sigma}$ commute, so the cross braiding is 1, while within each leg the braiding is that of A, resp. $\bar{A}$ by (i). □

BKLR's calculus is formulated for a braided C* tensor category realized in $\mathrm{End}_0(N)$ with N infinite, so the full braided subcategory $\mathcal{C} \boxtimes \mathcal{C}^{\mathrm{opp}}$ suffices; that it is all of $\mathrm{DHR}(A \otimes \bar{A})$ follows from [KLM, Cor. `c18`] and [Müger, Thm. `tprod`], but is not needed.

Theorem 7.5 (two-fold product of full centres)**.** *Let $\mathbf{A}, \mathbf{B}$ be simple Q-systems in $\mathcal{C}$, $Q = Z[\mathbf{A}]$, $Q' = Z[\mathbf{B}]$, localized as in Assumption 2.13. Then $D_Q \circledast D_{Q'} \cong \bigoplus_{\mathbf{m}} D_{\mathbf{m}}$ over the irreducible $\mathbf{A}$–$\mathbf{B}$-bimodules $\mathbf{m}$, where $D_{\mathbf{m}} := E_{\mathbf{m}} D_{Q''_h}$ is an irreducible defect.*

Proof. $Z[\mathbf{A}]$, $Z[\mathbf{B}]$ are Q-systems in $\mathrm{DHR}(\mathcal{A})$ (Lemma 7.4), irreducible by Proposition 2.11; by transportability one chooses localized representatives θ in K trivial on $\tilde{J}$, θ' in L, preserving the relations. By Theorem 6.4 and Lemma 6.3, $D_Q \circledast D_{Q'} \cong D_{Q''_h}$ with $Q''_h \simeq Z[\mathbf{A}] \times^{-} Z[\mathbf{B}]$. By Proposition 7.2 this is the sum over the minimal central projections, which by Theorem 2.10 (valid for either sign) are the $E_{\mathbf{m}}$. Each $E_{\mathbf{m}} D(I)$ is a factor. □

By Theorem 2.10, $\iota(N)' \cap E_{\mathbf{m}} M = E_{\mathbf{m}}(M' \cap M) = \mathbb{C} E_{\mathbf{m}}$, so each reduced Q-system $E_{\mathbf{m}}(Z[\mathbf{A}] \times Z[\mathbf{B}])$ is irreducible. Informally $D_{\mathbf{m}}$ is the defect "$\mathcal{A} \,|\, \mathcal{B}_2^{\mathbf{A}} \,|\, \mathbf{m} \,|\, \mathcal{B}_2^{\mathbf{B}} \,|\, \mathcal{A}$" with a phase boundary $\mathbf{m}$ between the full-centre extensions $\mathcal{B}_2^{\mathbf{A}}$, $\mathcal{B}_2^{\mathbf{B}}$ of $\mathcal{A}$; this identification with the phase-boundary defect of [BDH, Remark on three Q-systems] is not proved here (Section 9).

7.3 Corners

Lemma 7.6 (corners commute with fusion)**.** *Let $D = D_Q$, $E = D_{Q'}$ be as in Assumption 2.13 and let $z \in Z(D(I^{++}))$, $z' \in Z(E({}^{++}I))$ be central projections satisfying condition (N) of Lemma 7.1, so that zD, $z'E$ are defects. Then $z \in D(K)$, $z' \in E(L)$, zz' is a central projection of $(D \circledast E)(I)$, and $(zD) \circledast (z'E) \cong zz'(D \circledast E)$. Under the isomorphism of Theorem 6.4, zz' corresponds to $q^h \otimes q'$, where q, q' are the elements of $\mathrm{Hom}(1, Q)$, $\mathrm{Hom}(1, Q')$ corresponding to z, z' and $q^h = \alpha_h(q)$.*

Proof. z commutes with $\tilde{C} \subset D(I^{++})$, so $z \in D(I^{++}) \cap \tilde{C}' = D(K)$ (Assumption 2.13); likewise $z' \in E(L)$. $D(K)$ and $E(L)$ commute in $F = D(K) \vee E(L)$ and z, z' are central in them, so zz' is central in F. The vacuum sector of zD is $P_z H_1$ with P_z multiplication by z (proof of Lemma 7.1); P_z commutes with the right action of C, so by functoriality of Connes fusion $P_z H_1 \boxtimes_C P_{z'} H_2 = (P_z \boxtimes P_{z'})(H_1 \boxtimes_C H_2)$, and $(zD \circledast z'E)(I) = zD(K) \vee z'E(L)$ is the compression of F by zz'. The isomorphism Ψ of Theorem 6.4 is α_h on $\mathcal{A}(K)$, the identity on $\mathcal{A}(L)$, and $v \mapsto w'^{*} v''$, $v' \mapsto \alpha_h(w)^{*} v''$. Writing $z = \iota(a) v$ with $a \in \mathrm{Hom}(\theta, \mathrm{id})$ (Lemma 3.1) gives $\Psi(z) = \iota(\alpha_h(a) w'^{*}) v''$, and $\alpha_h(a) w'^{*} = \alpha_h(a) \theta^h(w'^{*})$ (as θ^h is trivial on the support of w') is the tensor product $a^h \otimes w'^{*} \in \mathrm{Hom}(\theta^h \theta', \mathrm{id})$, i.e. the element $q^h \otimes \eta_{Q'}$ (w'^{*} represents the unit). Likewise $\Psi(z')$ is $\eta_{Q^h} \otimes q'$, and the product is $q^h \otimes q'$ by Lemma 8.1. □

7.4 Associativity

Proposition 7.7 (associativity)**.** $(D_Q \circledast D_{Q'}) \circledast D_{Q''} \cong D_{(Q^{(h)} \times Q')^{(h')} \times Q''}$ *and* $D_Q \circledast (D_{Q'} \circledast D_{Q''}) \cong D_{Q \times (Q' \times Q'')}$ *for* Q*-systems* Q, Q', Q'' *localized from left to right; since the braided product is associative [BKLR],* $\circledast$ *is associative on* Q*-system defects.*

Proof. Apply Theorem 6.4 twice. Fusion is functorial for isomorphisms of defects, and $D_{Q''_h}$ is again of Q-system form with θ'' localized in an interval, so the hypotheses of the theorem hold in the new coordinates; the other bracketing is treated the same way. Equivalent Q-systems have isomorphic defects, so the h-relabelings do not matter (Lemma 6.3). □

The proof that the relabelings compose as claimed is by induction on the number of factors and is not written out.

For the three-fold product of full centres $Z[\mathbf{A}]$, $Z[\mathbf{B}]$, $Z[\mathbf{C}]$ the number of central summands is governed by the following dimension count. Let Z^a, Z^b, Z^c be the multiplicity matrices (modular invariants, so $[Z, S] = 0$ for the S-matrix of $\mathcal{C}$).

8 Products of full centres

Let $\mathbf{A}_0, \dots, \mathbf{A}_k$ be simple Q-systems in $\mathcal{C}$, $\mathcal{D}_i := \mathcal{D}_{\mathbf{A}_i}$ with F_i, I_i, $\mathcal{B}_i := {}_{\mathbf{A}_{i-1}}\mathcal{C}_{\mathbf{A}_i}$, and $\Theta_k := Z[\mathbf{A}_0] \times \cdots \times Z[\mathbf{A}_k]$ (any fixed choice of signs). For a tuple $t = (\mathbf{m}_1, \dots, \mathbf{m}_k)$, $\mathbf{m}_i \in \operatorname{Irr} \mathcal{B}_i$, put $X_t := \mathbf{m}_1 \otimes_{\mathbf{A}_1} \cdots \otimes_{\mathbf{A}_{k-1}} \mathbf{m}_k \in {}_{\mathbf{A}_0}\mathcal{C}_{\mathbf{A}_k}$.

Lemma 8.1 (sliding)**.** *Let* X, Y *be algebras in* $\mathcal{Z}$ *and* $q \in \operatorname{Hom}(1, X)$, $r \in \operatorname{Hom}(1, Y)$ *central. Then* $q \otimes \eta_Y$ *and* $\eta_X \otimes r$ *are central in* $X \times^{\pm} Y$*, the corner of* $X \times^{\pm} Y$ *by* $q \otimes \eta_Y$ *(if* q *is an idempotent) is the algebra* $(qX) \times^{\pm} Y$*, and* $(q \otimes \eta_Y)(\eta_X \otimes r) = q \otimes r$*. The same holds for any number of factors and any bracketing.*

Proof. Take the sign with $m_{X \times Y} = (m_X \otimes m_Y)(1_X \otimes c_{Y,X} \otimes 1_Y)$; the other sign replaces $c_{Y,X}$ by $c_{X,Y}^{-1}$ and the argument is the same. Naturality of the braiding with respect to morphisms out of 1, together with $c_{1,X} = c_{X,1} = \mathrm{id}$, gives $c_{Y,X}(\eta_Y \otimes 1_X) = 1_X \otimes \eta_Y$ and $c_{Y,X}(1_Y \otimes q) = q \otimes 1_Y$. Hence left multiplication by $q \otimes \eta_Y$ is $(m_X \otimes m_Y)(q \otimes 1_X \otimes \eta_Y \otimes 1_Y) = m_X(q \otimes 1_X) \otimes 1_Y$, and right multiplication is $(m_X \otimes m_Y)(1_X \otimes q \otimes 1_Y \otimes \eta_Y) = m_X(1_X \otimes q) \otimes 1_Y$; these agree because q is central in X. For $\eta_X \otimes r$ one uses $c_{Y,X}(r \otimes 1_X) = 1_X \otimes r$ and $c_{Y,X}(1_Y \otimes \eta_X) = \eta_X \otimes 1_Y$ in the same way. The formula for left multiplication shows that the corner is $(qX) \otimes Y$ with the restricted product $(m_{qX} \otimes m_Y)(1 \otimes c_{Y,qX} \otimes 1)$, i.e. $(qX) \times Y$, and the product formula is the case X-part q, Y-part r of the same computation. For more factors, apply this to $X \times (Y \times W)$ and $(X \times Y) \times W$ and use associativity [BKLR]. □

Theorem 8.2 (k-fold products)**.** *Assume (T1), (T2), (T7), and the multiplicativity of Proposition 3.4 (Remark 3.5). Then:*

(a) $\Theta_k \cong I_k(Y_k)$ *as algebras, where* $Y_k \cong \bigoplus_t \underline{\operatorname{End}}(X_t)$ *is a direct sum of algebras in* $\mathcal{D}_k$*, the internal Ends taken in the right* $\mathcal{D}_k$*-module category* ${}_{\mathbf{A}_0}\mathcal{C}_{\mathbf{A}_k}$*;*

(b) the relative commutant of the extension of Θ_k *is* $\bigoplus_t \operatorname{End}_{\mathbf{A}_0\text{-}\mathbf{A}_k}(X_t)$*;*

(c) the minimal central projections of the extension are labelled by the tuples t*, and the summand* t *has relative commutant* $\operatorname{End}(X_t)$*.*

Proof. Induction on k, with $Y_0 := 1 \in \mathcal{D}_0$ and $\Theta_0 = Z[\mathbf{A}_0] = I_0(1)$ (T1). Given (a) for $k-1$: $\Theta_k = \Theta_{k-1} \times Z[\mathbf{A}_k] \cong Z[\mathbf{A}_k] \times \Theta_{k-1} = I_k(1) \times I_{k-1}(Y_{k-1}) \cong I_k\big(F_k I_{k-1}(Y_{k-1})\big)$ by (T2) and

Lemma 3.3. By Proposition 3.4, $Y_k := F_k I_{k-1}(Y_{k-1}) = \bigoplus_{t',\mathbf{n}} \underline{\mathrm{Hom}}_{\mathcal{B}_k}(\mathbf{n}, \underline{\mathrm{End}}(X_{t'}) \otimes \mathbf{n})$, t' a $(k-1)$-tuple and $\mathbf{n} \in \mathrm{Irr}\,\mathcal{B}_k$ (the sum over t' splits because Y_{k-1} is a direct sum of algebras). Writing $\underline{\mathrm{End}}(X_{t'}) = \bar{X}_{t'} \otimes_{\mathbf{A}_0} X_{t'} \in \mathcal{D}_{k-1}$ and $\underline{\mathrm{Hom}}_{\mathcal{B}_k}(\mathbf{n}, U \otimes \mathbf{n}) = \bar{\mathbf{n}} \otimes_{\mathbf{A}_{k-1}} U \otimes_{\mathbf{A}_{k-1}} \mathbf{n}$, the summand is $\overline{X_{t'} \otimes \mathbf{n}} \otimes_{\mathbf{A}_0} (X_{t'} \otimes \mathbf{n}) = \underline{\mathrm{End}}(X_{(t',\mathbf{n})})$ in $\mathcal{D}_k$, and composition matches composition. This is (a). (b): $\mathrm{Hom}(1, I_k(Y_k)) = \mathrm{Hom}(1, Y_k) = \bigoplus_t \mathrm{End}(X_t)$ as algebras (Lemma 3.3, last sentence, and Lemma 3.1). (c): by Lemma 3.2 and Proposition 3.6 (whose proof only used $\Theta \cong I(A)$ with A an algebra in $\mathcal{D}$), $\dim Z =$ the number of indecomposable summands of $\mathrm{Mod}_{Y_k}(\mathcal{D}_k) = \bigoplus_t \mathrm{Mod}_{\underline{\mathrm{End}}(X_t)}(\mathcal{D}_k)$, and each $\mathrm{Mod}_{\underline{\mathrm{End}}(X_t)}(\mathcal{D}_k)$ is the submodule category of the indecomposable ${}_{\mathbf{A}_0}\mathcal{C}_{\mathbf{A}_k}$ generated by $X_t \neq 0$, i.e. all of it. The unit of the summand $I_k(\underline{\mathrm{End}}(X_t))$ is the corresponding minimal central projection, and its relative commutant is $\mathrm{End}(X_t)$ by (b). □

For $k = 1$, Theorem 8.2 recovers the labelling of the centre of $Z[\mathbf{A}] \times Z[\mathbf{B}]$ by bimodules (Theorem 2.10); **we fix this labelling**: $e_{\mathbf{m}}$ is the unit of the summand $I_{\mathbf{B}}(\underline{\mathrm{End}}(\mathbf{m}))$ of $Z[\mathbf{A}] \times Z[\mathbf{B}] \cong I_{\mathbf{B}}(Y_1)$, and $D_{\mathbf{m}} := e_{\mathbf{m}}(D_{Z[\mathbf{A}]} \circledast D_{Z[\mathbf{B}]})$. (Whether it coincides with the bijection chosen in [BKLR] is not needed.) For $k = 2$ the next proposition identifies the central projections of the three-fold product in terms of the two-fold ones.

Proposition 8.3 (labels)**.** *In* $\Theta_2 = Z[\mathbf{A}] \times Z[\mathbf{B}] \times Z[\mathbf{C}]$ *put* $f_{\mathbf{m}_1\mathbf{m}_2} := (e_{\mathbf{m}_1} \otimes \eta)(\eta \otimes e_{\mathbf{m}_2})$*. Under the hypotheses of Theorem 8.2,* $f_{\mathbf{m}_1\mathbf{m}_2}$ *is the minimal central projection with label* $(\mathbf{m}_1, \mathbf{m}_2)$*; its relative commutant is* $\mathrm{End}_{\mathbf{A}\text{-}\mathbf{C}}(\mathbf{m}_1 \otimes_{\mathbf{B}} \mathbf{m}_2)$.

Proof. In the induction step of Theorem 8.2, $\Theta_2 \cong Z[\mathbf{C}] \times \Theta_1 \cong I_{\mathbf{C}}(F_{\mathbf{C}}\Theta_1)$, and on $\mathrm{Hom}(1, \cdot)$ the projection isomorphism sends $\eta \otimes q \mapsto F_{\mathbf{C}}(q)$ (it is adjoint to $\varepsilon \otimes 1$). *The factor* $e_{\mathbf{m}_1} \otimes \eta$, moved in front of $Z[\mathbf{C}]$ by (T2), goes to $F_{\mathbf{C}}I_{\mathbf{B}}$ applied to the unit of $\underline{\mathrm{End}}(\mathbf{m}_1) \subset Y_1$, which by naturality and additivity of Proposition 3.4 in V is the projection onto $\bigoplus_{\mathbf{n}} \underline{\mathrm{Hom}}(\mathbf{n}, \underline{\mathrm{End}}(\mathbf{m}_1) \otimes \mathbf{n})$. *The factor* $\eta \otimes e_{\mathbf{m}_2}$: by the same recipe for $(\mathbf{B}, \mathbf{C})$, $Z[\mathbf{B}] \times Z[\mathbf{C}] \cong I_{\mathbf{C}}(F_{\mathbf{C}}Z[\mathbf{B}]) = I_{\mathbf{C}}(\bigoplus_{\mathbf{n}} \underline{\mathrm{End}}(\mathbf{n}))$ and $e_{\mathbf{m}_2}$ is the unit of the summand $\mathbf{n} = \mathbf{m}_2$. The algebra map $Z[\mathbf{B}] \to \Theta_1$, $y \mapsto \eta \otimes y$, is $I_{\mathbf{B}}$ of the unit $1 \to Y_1$; by naturality of the projection isomorphism and of Proposition 3.4 in V, $\eta \otimes e_{\mathbf{m}_2}$ goes to the image of $1_{\mathbf{m}_2}$ under $\underline{\mathrm{End}}(\mathbf{m}_2) \to \underline{\mathrm{Hom}}(\mathbf{m}_2, Y_1 \otimes \mathbf{m}_2)$, the unit of the summand $\mathbf{n} = \mathbf{m}_2$. *Product:* $f_{\mathbf{m}_1\mathbf{m}_2}$ is the unit of $\underline{\mathrm{Hom}}(\mathbf{m}_2, \underline{\mathrm{End}}(\mathbf{m}_1) \otimes \mathbf{m}_2) = \underline{\mathrm{End}}(\mathbf{m}_1 \otimes_{\mathbf{B}} \mathbf{m}_2)$, the summand $t = (\mathbf{m}_1, \mathbf{m}_2)$ in Theorem 8.2. □

9 Proof of Remark 1.1

9.1 (E1)

(E1) is Theorem 1.2: Theorem 6.4 identifies $D_Q \circledast D_{Q'}$ with $D_{Q^h \times Q'}$, Proposition 6.6 shows it is a defect, and Proposition 7.7 gives associativity, matching associativity of the braided product (T2).

9.2 Three lemmas

Lemma 9.1 (unit summand)**.** *Let* L *be a commutative* Q*-system in* $\mathcal{Z}$ *with* $\mathrm{Hom}(1, L) = \mathbb{C}\eta_L$ *(e.g.* $L = Z[\mathbf{B}]$ *for simple* $\mathbf{B}$*, by [BKLR] commutativity of the full centre and Proposition 2.11), and* $P := L \times^{\pm} L$*. Then:*

(a) the multiplication $m\colon P \to L$ *is a unital algebra homomorphism;*

(b) *there is exactly one minimal central idempotent* $e_0 \in \mathrm{Hom}(1,P)$ *with* $m\circ e_0 \neq 0$*; it satisfies* $m\circ e_0 = \eta_L$*, and* $m\circ e = 0$ *for every other minimal central idempotent* e*;*
(c) $m|_{e_0P}\colon e_0P \to L$ *is an isomorphism of algebras.*

Proof. (a) With $m^{(k)}\colon L^{\otimes k} \to L$ the iterated product, $m\circ m_P = m\circ(m\otimes m)(1\otimes c^{\pm}\otimes 1) = m^{(3)}(1\otimes mc^{\pm}\otimes 1) = m^{(3)}(1\otimes m\otimes 1) = m^{(4)} = m\circ(m\otimes m)$, using commutativity $m\circ c_{L,L} = m$ and hence also $m\circ c_{L,L}^{-1} = m$. Unitality: $m(\eta_L\otimes\eta_L) = \eta_L$. (b) The minimal central idempotents e_k of P satisfy $\sum_k e_k = \eta_P$ and $e_k e_l = \delta_{kl}e_k$. By (a), each $m\circ e_k \in \mathrm{Hom}(1,L) = \mathbb{C}\eta_L$ is an idempotent of $\mathrm{Hom}(1,L)\cong\mathbb{C}$, so it is 0 or η_L; the images are orthogonal and sum to $m\circ\eta_P = \eta_L$. Exactly one is non-zero. (c) *Surjective:* for the morphism $s := e_0\cdot(\eta_L\otimes 1_L)\colon L\to e_0P$ (left multiplication by e_0 after inserting the unit), (a) and (b) give $m\circ s = m(e_0)\cdot m(\eta_L\otimes 1_L) = 1_L$. *Injective:* $\ker(m|_{e_0P})$ is a P–P-sub-bimodule of e_0P, since m is an algebra homomorphism; the category of P–P-bimodules is semisimple (P is a Q-system, hence separable), so the kernel is a direct summand, whose projection is a bimodule endomorphism of e_0P. Bimodule endomorphisms of P are its centre (Lemma 3.2), and e_0 is minimal, so $\mathrm{End}_{P\text{-}P}(e_0P) = \mathbb{C}e_0$; thus the kernel is 0 or e_0P, and it is not e_0P because $m(e_0)\neq 0$. A morphism in a semisimple category that is injective and surjective is an isomorphism. □

Lemma 9.2 (the unit summand is the Casimir)**.** *In the situation of Lemma 9.1, let* $c := d_L^{-1}\, x\circ\eta_L \in \mathrm{Hom}(1, L\otimes L)$ *(the normalized Casimir of the Frobenius algebra* L*). Then left and right multiplication by* c *in* $P = L\times^{\pm} L$ *are both equal to* $d_L^{-1}\, x\circ m$*, and* $c = E_{\mathbf{B}} := e_0$*. In particular left multiplication by* e_0 *is the orthogonal projection* $d_L^{-1}x\circ m$*.*

Proof. Take the sign with $m_P = (m\otimes m)(1\otimes c_{L,L}\otimes 1)$. Writing the braiding of the object L with the pair of legs of the Casimir as $c_{L\otimes L,L} = (c_{L,L}\otimes 1)(1\otimes c_{L,L})$ and using naturality, $c_{L\otimes L,L}(x\eta\otimes 1) = 1\otimes x\eta$, one gets $(1\otimes c_{L,L})(x\eta\otimes 1_L) = (c_{L,L}^{-1}\otimes 1)(1_L\otimes x\eta)$. Hence

$$(m\otimes 1)(1\otimes c_{L,L})(x\eta\otimes 1_L) = (m\, c_{L,L}^{-1}\otimes 1)(1\otimes x\eta) = (m\otimes 1)(1\otimes x\eta) = x,$$

by commutativity ($m\, c^{-1} = m$) and the Frobenius property. Therefore $c\star(a\otimes b) = d^{-1}(1\otimes m)(x\otimes 1)(a\otimes b) = d^{-1}x\, m(a\otimes b)$, again by Frobenius. The other sign and right multiplication are the mirror computations. So c is central, $c\star c = d^{-1}x\, m(c) = d^{-1}x\eta = c$ (as $mx = d$), and left multiplication by c is $d^{-1}xm$, an orthogonal projection (as $x = m^*$) with kernel $\ker m$. By Lemma 9.1 $\ker m = (1-e_0)P$; two central idempotents generating the same ideal are equal, so $c = e_0$. □

Lemma 9.2 also gives Lemma 9.1(c) directly: $e_0P = d^{-1}x\, m(P) = x(L)\cong L$.

Lemma 9.3 (algebra isomorphisms of haploid Q-systems)**.** *Let* P, L *be* Q*-systems in* $\mathcal{Z}$ *with* $\mathrm{Hom}(1,L) = \mathbb{C}$*, of the same dimension, and* $n\colon P\to L$ *an isomorphism of unital algebras. Then there is a unitary* $u\colon P\to L$ *which is an isomorphism of* Q*-systems. In particular the defects* D_P *and* D_L *are isomorphic.*

Proof. Counit. $\mathrm{Hom}(P,1)\cong\mathrm{Hom}(1,P)^*$ is one-dimensional (as $P\cong L$), so $\varepsilon_L\circ n = \lambda\varepsilon_P$; evaluating on η_P and using $n\eta_P = \eta_L$ and $\varepsilon(\eta) = d$ (standard normalization, equal dimensions) gives $\lambda = 1$. *Coalgebra.* For a Frobenius algebra the coproduct is determined by (m,η,ε) (it is the unique map making $\varepsilon\circ m$ a Frobenius pairing), so n also intertwines the coproducts; i.e. n is an isomorphism of Frobenius algebras, and for Q-systems the coproduct is $x = m^*$. *Adjoint.* Taking adjoints of $n\, m_P = m_L(n\otimes n)$, $n\eta_P = \eta_L$, $\varepsilon_L n = \varepsilon_P$ shows that n^* intertwines

coproducts, counits and units, hence (same determination argument) is also an isomorphism of Frobenius algebras $L \to P$. So $a := n^*n$ is a positive Frobenius automorphism of P. *Polar part.* Write $a = \sum_i \lambda_i p_i$ (spectral projections $p_i \in \mathrm{End}(P)$, $\lambda_i > 0$). The identities $a^t m = m(a^t \otimes a^t)$, $a^t \eta = \eta$ hold for all $t \in \mathbb{Z}$; they are finite linear relations among the functions $t \mapsto \lambda_i^t, (\lambda_j\lambda_k)^t$, so they hold for all real t (exponentials with distinct bases are linearly independent). Hence $a^{-1/2}$ is a Frobenius automorphism and $u := n\, a^{-1/2}$ is a unitary isomorphism of Q-systems. *Defects.* $u \in \mathrm{Hom}(\theta_P, \theta_L)$ intertwines two endomorphisms localized in $[0,1]$, so $u \in \mathcal{A}([0,1])$ by Haag duality, and the Q-systems are equivalent in the sense of Definition 2.4; equivalent Q-systems have isomorphic defects ($v \mapsto uv$). □

Proposition 9.4 (the pair projections)**.** *Let $\mathbf{A}, \mathbf{B}, \mathbf{C}$ be simple, let $e_{\mathbf{m}_1}$ ($\mathbf{m}_1 \in \mathrm{Irr}({}_{\mathbf{A}}\mathcal{C}_{\mathbf{B}})$) be the minimal central idempotents of $Z[\mathbf{A}] \times Z[\mathbf{B}]$ and $e_{\mathbf{m}_2}$ those of $Z[\mathbf{B}] \times Z[\mathbf{C}]$ (Theorem 2.10), and in $\Theta = Z[\mathbf{A}] \times Z[\mathbf{B}] \times Z[\mathbf{C}]$ put*

$$f_{\mathbf{m}_1\mathbf{m}_2} := (e_{\mathbf{m}_1} \otimes \eta_{Z[\mathbf{C}]})\,(\eta_{Z[\mathbf{A}]} \otimes e_{\mathbf{m}_2}).$$

Then the $f_{\mathbf{m}_1\mathbf{m}_2}$ are pairwise orthogonal central projections of Θ with $\sum f_{\mathbf{m}_1\mathbf{m}_2} = 1$, and with the normalized states $\tau_P := d_P^{-1}\varepsilon_P$,

$$\tau_\Theta(f_{\mathbf{m}_1\mathbf{m}_2}) = \tau(e_{\mathbf{m}_1})\,\tau(e_{\mathbf{m}_2}) > 0.$$

In particular every $f_{\mathbf{m}_1\mathbf{m}_2}$ is non-zero, $\dim(M' \cap M) \geq \#\mathrm{Irr}({}_{\mathbf{A}}\mathcal{C}_{\mathbf{B}}) \cdot \#\mathrm{Irr}({}_{\mathbf{B}}\mathcal{C}_{\mathbf{C}})$. By Proposition 8.3 they are exactly the minimal central projections.

Proof. Both factors are central idempotents by Lemma 8.1, so they commute and their product is a central idempotent (a projection, since central idempotents in the C*-algebra $M' \cap M$ are projections). Orthogonality and $\sum = 1$ follow from those of the e's. By Lemma 8.1, $f_{\mathbf{m}_1\mathbf{m}_2} = (1 \otimes m_{Z[\mathbf{B}]} \otimes 1)(e_{\mathbf{m}_1} \otimes e_{\mathbf{m}_2})$ as an element of $\mathrm{Hom}(1, Z[\mathbf{A}] \otimes Z[\mathbf{B}] \otimes Z[\mathbf{C}])$. Apply $\varepsilon_\Theta = \varepsilon_{\mathbf{A}} \otimes \varepsilon_{\mathbf{B}} \otimes \varepsilon_{\mathbf{C}}$ (counits of the full centres): with $u := (\varepsilon \otimes 1)e_{\mathbf{m}_1}$ and $u' := (1 \otimes \varepsilon)e_{\mathbf{m}_2}$ in $\mathrm{Hom}(1, Z[\mathbf{B}]) = \mathbb{C}\eta$, say $u = \lambda\eta$, $u' = \lambda'\eta$, one gets $\varepsilon_\Theta(f) = \varepsilon_{Z[\mathbf{B}]}(m(u \otimes u')) = \lambda\lambda' d_{Z[\mathbf{B}]}$, while $\varepsilon(e_{\mathbf{m}_1}) = \lambda d_{Z[\mathbf{B}]}$ and $\varepsilon(e_{\mathbf{m}_2}) = \lambda' d_{Z[\mathbf{B}]}$. Hence $\varepsilon_\Theta(f) = \varepsilon(e_{\mathbf{m}_1})\varepsilon(e_{\mathbf{m}_2})/d_{Z[\mathbf{B}]}$, which is the stated identity because $d_\Theta = d_{Z[\mathbf{A}]}d_{Z[\mathbf{B}]}d_{Z[\mathbf{C}]}$. On $\iota(N)' \cap M$, τ is the restriction of the conditional expectation E (since $E(\iota(a)v) = d^{-1}aw$ and $aw \in \mathbb{C}$), which is faithful (L1); so $\tau(e) > 0$ for every non-zero projection, and $\tau(f) > 0$. The last two assertions are counting. □

9.3 The summand over the phase B

Theorem 9.5 (the boundary-condition summand)**.** *Let $\mathbf{A}, \mathbf{B}, \mathbf{C}$ be simple Q-systems in the modular category $\mathcal{C}$, $\mathbf{m}_1 \in \mathrm{Irr}({}_{\mathbf{A}}\mathcal{C}_{\mathbf{B}})$, $\mathbf{m}_2 \in \mathrm{Irr}({}_{\mathbf{B}}\mathcal{C}_{\mathbf{C}})$, and $D_{\mathbf{m}_1} = e_{\mathbf{m}_1}(D_{Z[\mathbf{A}]} \circledast D_{Z[\mathbf{B}]})$, $D_{\mathbf{m}_2} = e_{\mathbf{m}_2}(D_{Z[\mathbf{B}]} \circledast D_{Z[\mathbf{C}]})$ the irreducible defects of Theorem 7.5. Then*

$$D_{\mathbf{m}_1} \circledast D_{\mathbf{m}_2} \;\cong\; D_{(\mathbf{m}_1,\mathbf{m}_2)} \;\oplus\; D', \qquad D_{(\mathbf{m}_1,\mathbf{m}_2)} := f_{\mathbf{m}_1\mathbf{m}_2}\big(D_{Z[\mathbf{A}]} \circledast D_{Z[\mathbf{B}]} \circledast D_{Z[\mathbf{C}]}\big) \neq 0,$$

where $D' = \bigoplus_{e \neq E_{\mathbf{B}}}(e_{\mathbf{m}_1} \otimes e \otimes e_{\mathbf{m}_2})(D_{Z[\mathbf{A}]} \circledast D_{Z[\mathbf{B}]} \circledast D_{Z[\mathbf{B}]} \circledast D_{Z[\mathbf{C}]})$, the sum over the other minimal central idempotents e of $Z[\mathbf{B}] \times Z[\mathbf{B}]$. By Proposition 8.3, $D_{(\mathbf{m}_1,\mathbf{m}_2)}$ is irreducible with relative commutant $\mathrm{End}(\mathbf{m}_1 \otimes_{\mathbf{B}} \mathbf{m}_2)$.

Proof. All Möbius relabelings h are absorbed by Lemma 6.3 and Proposition 7.7; I suppress them. By Lemma 7.6 and Theorem 1.2, $D_{\mathbf{m}_1} \circledast D_{\mathbf{m}_2} \cong p\, D_{\Theta_4}$ with $\Theta_4 = (Z[\mathbf{A}] \times Z[\mathbf{B}]) \times (Z[\mathbf{B}] \times Z[\mathbf{C}])$ and $p = e_{\mathbf{m}_1} \otimes e_{\mathbf{m}_2}$, a central projection of Θ_4 by Lemma 8.1. Let $P = Z[\mathbf{B}] \times Z[\mathbf{B}]$;

by associativity $\Theta_4 = Z[\mathbf{A}] \times P \times Z[\mathbf{C}]$, and for each minimal central idempotent e of P the element $q_e := \eta \otimes e \otimes \eta$ is central in Θ_4 (Lemma 8.1), with $\sum_e q_e = 1$. So $pD_{\Theta_4} = \bigoplus_e p\, q_e D_{\Theta_4}$ (Lemma 7.1). For $e = E_{\mathbf{B}}$: by Lemma 8.1 the corner $q_{E_{\mathbf{B}}}\Theta_4$ is $Z[\mathbf{A}] \times E_{\mathbf{B}}P \times Z[\mathbf{C}]$, and $\psi := 1 \otimes m|_{E_{\mathbf{B}}P} \otimes 1$ is an algebra isomorphism onto $\Theta = Z[\mathbf{A}] \times Z[\mathbf{B}] \times Z[\mathbf{C}]$ (Lemma 9.1(c); the braided product of algebra isomorphisms is an algebra isomorphism, by naturality of the braiding). Moreover ψ extends to the algebra homomorphism $1 \otimes m \otimes 1 \colon \Theta_4 \to \Theta$, under which $e_{\mathbf{m}_1} \otimes \eta \otimes \eta \mapsto e_{\mathbf{m}_1} \otimes \eta$ and $\eta \otimes \eta \otimes e_{\mathbf{m}_2} \mapsto \eta \otimes e_{\mathbf{m}_2}$ (because $m(\eta \otimes 1) = m(1 \otimes \eta) = 1$), so $\psi(pq_{E_{\mathbf{B}}}) = f_{\mathbf{m}_1\mathbf{m}_2}$. By Lemma 9.3 the corner defect $pq_{E_{\mathbf{B}}}D_{\Theta_4}$ is isomorphic to $f_{\mathbf{m}_1\mathbf{m}_2}D_\Theta = D_{(\mathbf{m}_1,\mathbf{m}_2)}$, which is non-zero by Proposition 9.4. The remaining terms form D'. □

Corollary 9.6 (complete decomposition of $D_{\mathbf{m}_1} \circledast D_{\mathbf{m}_2}$)**.** *Under the hypotheses of Theorem 8.2,*

$$D_{\mathbf{m}_1} \circledast D_{\mathbf{m}_2} \;\cong \bigoplus_{\mathbf{m}' \in \mathrm{Irr}({}_{\mathbf{B}}\mathcal{C}_{\mathbf{B}})} D_{(\mathbf{m}_1,\mathbf{m}',\mathbf{m}_2)},$$

where $D_{(\mathbf{m}_1,\mathbf{m}',\mathbf{m}_2)}$ is the irreducible summand of the four-fold fusion $D_{Z[\mathbf{A}]} \circledast D_{Z[\mathbf{B}]} \circledast D_{Z[\mathbf{B}]} \circledast D_{Z[\mathbf{C}]}$ with label $(\mathbf{m}_1, \mathbf{m}', \mathbf{m}_2)$ and relative commutant $\mathrm{End}_{\mathbf{A}\text{-}\mathbf{C}}(\mathbf{m}_1 \otimes_{\mathbf{B}} \mathbf{m}' \otimes_{\mathbf{B}} \mathbf{m}_2)$. The term $\mathbf{m}' = \mathbf{B}$ is $D_{(\mathbf{m}_1,\mathbf{m}_2)}$ of Theorem 9.5, consistently with $\mathbf{m}_1 \otimes_{\mathbf{B}} \mathbf{B} \otimes_{\mathbf{B}} \mathbf{m}_2 = \mathbf{m}_1 \otimes_{\mathbf{B}} \mathbf{m}_2$. In particular $D_{\mathbf{m}_1} \circledast D_{\mathbf{m}_2}$ has exactly $\#\mathrm{Irr}({}_{\mathbf{B}}\mathcal{C}_{\mathbf{B}})$ irreducible summands.

Proof. By the proof of Theorem 9.5, $D_{\mathbf{m}_1} \circledast D_{\mathbf{m}_2} \cong pD_{\Theta_4}$ with $p = e_{\mathbf{m}_1} \otimes e_{\mathbf{m}_2}$. Theorem 8.2 with $(\mathbf{A}_0, \ldots, \mathbf{A}_3) = (\mathbf{A}, \mathbf{B}, \mathbf{B}, \mathbf{C})$ decomposes Θ_4 by triples. The labels of p are tracked exactly as in Proposition 8.3: $e_{\mathbf{m}_1} \otimes \eta \otimes \eta$ and $\eta \otimes \eta \otimes e_{\mathbf{m}_2}$ project onto the triples with first, resp. last, entry $\mathbf{m}_1$, resp. $\mathbf{m}_2$ (the first by the induction, which starts with $Z[\mathbf{A}] \times Z[\mathbf{B}]$; the last by naturality in Y_2 of the final induction step, as in Proposition 8.3). So p is the sum of the minimal central projections $(\mathbf{m}_1, \mathbf{m}', \mathbf{m}_2)$, $\mathbf{m}'$ arbitrary. The identification of the term $\mathbf{m}' = \mathbf{B}$ is Lemma 9.1 via Theorem 9.5 (with the label of $E_{\mathbf{B}}$ being $\mathbf{B}$: $E_{\mathbf{B}}$ is the summand on which $m \colon Z[\mathbf{B}] \times Z[\mathbf{B}] \to Z[\mathbf{B}]$ is an isomorphism, and its relative commutant in the three-fold product is $\mathrm{End}(\mathbf{m}_1 \otimes \mathbf{m}_2)$ while that of the summand $\mathbf{m}'$ is $\mathrm{End}(\mathbf{m}_1 \otimes \mathbf{m}' \otimes \mathbf{m}_2)$; for the identification of labels it suffices that under Theorem 8.2(a) for $k = 1$ the unit summand $I_{\mathbf{B}}(\underline{\mathrm{End}}(\mathbf{m}'))$ with $\mathbf{m}' = \mathbf{B}$ is $I_{\mathbf{B}}(1) = Z[\mathbf{B}]$, the target of m). □

9.4 (E2)

Theorem 9.7 (Remark 1.1, (E2))**.** *Let $\mathbf{A}, \mathbf{B}, \mathbf{C}$ be simple Q-systems in $\mathcal{C}$, $\mathbf{m}_1 \in \mathrm{Irr}({}_{\mathbf{A}}\mathcal{C}_{\mathbf{B}})$, $\mathbf{m}_2 \in \mathrm{Irr}({}_{\mathbf{B}}\mathcal{C}_{\mathbf{C}})$.*

(i) $D_{\mathbf{m}_1} \circledast D_{\mathbf{m}_2}$ contains the summand $D_{(\mathbf{m}_1,\mathbf{m}_2)} = f_{\mathbf{m}_1\mathbf{m}_2}(D_{Z[\mathbf{A}]} \circledast D_{Z[\mathbf{B}]} \circledast D_{Z[\mathbf{C}]})$, which is non-zero; the $D_{(\mathbf{m}_1,\mathbf{m}_2)}$ are orthogonal and exhaust the three-fold fusion. (Theorem 9.5; uses Theorem 1.2, [BKLR], (T1), (T2), L1.)

(ii) $D_{(\mathbf{m}_1,\mathbf{m}_2)}$ is irreducible and its relative commutant is $\mathrm{End}_{\mathbf{A}\text{-}\mathbf{C}}(\mathbf{m}_1 \otimes_{\mathbf{B}} \mathbf{m}_2) \cong \bigoplus_{\mathbf{m}} M_{N_{\mathbf{m}_1\mathbf{m}_2}^{\mathbf{m}}}(\mathbb{C})$. (Proposition 8.3; uses in addition (T3)–(T7) and Remark 3.5.)

(iii) $D_{\mathbf{m}_1} \circledast D_{\mathbf{m}_2} \cong \bigoplus_{\mathbf{m}'} D_{(\mathbf{m}_1,\mathbf{m}',\mathbf{m}_2)}$ with irreducible summands of relative commutant $\mathrm{End}(\mathbf{m}_1 \otimes_{\mathbf{B}} \mathbf{m}' \otimes_{\mathbf{B}} \mathbf{m}_2)$, and the term $\mathbf{m}' = \mathbf{B}$ is $D_{(\mathbf{m}_1,\mathbf{m}_2)}$. (Corollary 9.6; same inputs as (ii).)

Proof. Collect the cited results. □

9.5 Interpretation

Item (ii) is the precise form of "the juxtaposition of $\mathbf{m}_1$ and $\mathbf{m}_2$ is described by $\mathbf{m}_1 \otimes_{\mathbf{B}} \mathbf{m}_2$": the summand of the juxtaposed boundary conditions sitting over the phase $\mathbf{B}$ has as its endomorphisms exactly those of the bimodule $\mathbf{m}_1 \otimes_{\mathbf{B}} \mathbf{m}_2$, so it splits, up to equivalence of defects, as $\bigoplus_{\mathbf{m}} N^{\mathbf{m}}_{\mathbf{m}_1\mathbf{m}_2}$ copies of the pieces labelled $\mathbf{m}$. When some $N^{\mathbf{m}}_{\mathbf{m}_1\mathbf{m}_2} \geq 2$ this relative commutant is non-commutative, so $D_{(\mathbf{m}_1,\mathbf{m}_2)}$ is irreducible but not isomorphic to any $D_{\mathbf{m}}$ (whose relative commutant is $\mathbb{C}$); "described by" is therefore a statement about endomorphism algebras, not an isomorphism $D_{(\mathbf{m}_1,\mathbf{m}_2)} \cong \bigoplus N D_{\mathbf{m}}$. Item (iii) explains the role of the middle phase: Connes fusion over $\mathcal{A}$ inserts the defect $D_{Z[\mathbf{B}]} \circledast D_{Z[\mathbf{B}]} = \bigoplus_{\mathbf{m}'} D_{\mathbf{m}'}$, and only $\mathbf{m}' = \mathbf{B}$ is the identity of the phase $\mathbf{B}$ (Lemma 9.1).

9.6 What remains to be done

(a) *Unwritten diagram chases:* the multiplicativity in Remark 3.5; the module compatibility in Proposition 3.6; the induction on Möbius relabelings in Proposition 7.7.

(b) *Independent reading* of the analytic core of Theorem 1.2, Lemmas 5.3 and 5.4, and of Lemmas 9.1, 9.3 and Proposition 9.4, which carry the new content of (E2).

(c) *Not addressed:* whether $D_{\mathbf{m}}$ is the phase-boundary defect between the extended nets $\mathcal{A}_{\mathbf{A}}$, $\mathcal{A}_{\mathbf{B}}$ in the sense of [BDH]; the results here do not need this identification.

10 Applications

The applications of these results to two-dimensional conformal quantum field theory are in the companion notes `connes_fusion_applications`: the time-zero dictionary with the boundary nets of [BKLR2], fusion through an intermediate phase, defect–boundary fusion for hard boundaries, sectors, junctions, crossings of transparent boundaries, corners, lightlike boundaries, null interfaces, and hard boundaries meeting at a point. References to it are marked A (for example A 7.3).

A Numerical checks

All proofs above are by hand. As a test of the categorical statements I computed in group-theoretical models: $\mathcal{C}$ replaced by Vec_G (or $\mathrm{Rep}\, G$), $\mathcal{Z} = Z(\mathrm{Vec}_G)$, and $Z[\mathbb{C}[H]]$ for subgroups $H \leq G$ given explicitly by [Davydov, Prop. 9.5]: basis (xfx^{-1}, xH), $f \in H$, graded by the first entry, G acting by conjugation and translation, product $(g,c)(g',c') = \delta_{cc'}(gg',c)$. Bimodule fusion rules were computed independently from characters of $K \times K'$-equivariant bundles on G. $G = S_3$ (all 4 conjugacy classes of subgroups) and $G = A_4$ (all 5); "both signs" means both braided products.

statement tested	cases	result
three-fold centre dimension $= \#\mathrm{Irr}({}_{\mathbf{A}}\mathcal{C}_{\mathbf{B}}) \cdot \#\mathrm{Irr}({}_{\mathbf{B}}\mathcal{C}_{\mathbf{C}})$ (Thm. 8.2, $k=2$)	64 + 125 triples, both signs	378/378
three-fold blocks: (dimension, number of simple summands) per central summand vs. $\mathrm{End}(\mathbf{m}_1 \otimes \mathbf{m}_2)$, as multisets	64 + 125 triples	189/189 (15 with a multiplicity 2)
pair projections $f_{\mathbf{m}_1\mathbf{m}_2}$ non-zero, central, idempotent, minimal; trace identity of Prop. 9.4	64 + 125 triples, both signs	378/378
labelled blocks: one bijection idempotents↔bimodules matching every $f_{\mathbf{m}_1\mathbf{m}_2}$ (Prop. 8.3)	7 ($\mathbf{A}=\mathbf{B}=\mathbf{C}$)	7/7 (weak test)
Y-junction corner $p\Theta p \cong e_{\mathbf{m}}(L\times L)$ (Prop. A 6.1)	1 (A_4, $N=2$)	1/1
unit summand: unique e with $m(e)\neq 0$, $m\|_{e(L\times L)}$ bijective (Lemma 9.1)	9 subgroups, both signs	18/18
Casimir $d^{-1}x\eta$ equals the unit summand e_0 (Lemma 9.2)	9 subgroups, both signs	18/18
four-fold centre dimension $= \prod \#\mathrm{Irr}$ (Thm. 8.2, $k=3$)	256 four-tuples (S_3), both signs	512/512
four-fold blocks vs. $\mathrm{End}(\mathbf{m}_1 \otimes \mathbf{m}' \otimes \mathbf{m}_2)$	6 four-tuples (S_3)	6/6

In addition the algebraic identities of Lemma 4.1 were tested in a Cuntz-algebra model and Lemma 5.4 in a type I model. Scripts: `threefold_subgroups.py`, `sweep.py`, `bimod_fusion.py`, `blocks_compare.py`, `pair_projections.py`, `label_match.py`, `unit_alg.py`, `nfold_centre.py`, `fourfold_blocks.py`, `casimir_check.py`, `junction_check.py`.

References

[1] A. Bartels, C. Douglas, A. Henriques, *Conformal nets III: fusion of defects*, arXiv:1310.8263; *Conformal nets V: dualizability*, arXiv:1905.03393.

[2] M. Bischoff, Y. Kawahigashi, R. Longo, K.-H. Rehren, *Tensor categories and endomorphisms of von Neumann algebras (with applications to quantum field theory)*, SpringerBriefs 2015, arXiv:1407.4793.

[3] M. Bischoff, Y. Kawahigashi, R. Longo, K.-H. Rehren, *Phase boundaries in algebraic conformal QFT*, Commun. Math. Phys. 342 (2016), arXiv:1405.7863.

[4] A. Davydov, *Centre of an algebra*, arXiv:0908.1250.

[5] A. Davydov, M. Müger, D. Nikshych, V. Ostrik, *The Witt group of non-degenerate braided fusion categories*, arXiv:1009.2117.

[6] V. Ostrik, *Module categories, weak Hopf algebras and modular invariants*, Transform. Groups 8 (2003), arXiv:math/0111139.

[7] P. Etingof, S. Gelaki, D. Nikshych, V. Ostrik, *Tensor categories*, AMS 2015.

[8] P. Etingof, D. Nikshych, V. Ostrik, *Fusion categories and homotopy theory*, arXiv:0909.3140.

[9] Y. Kawahigashi, R. Longo, M. Müger, *Multi-interval subfactors and modularity of representations in conformal field theory*, CMP 219 (2001).

[10] H. Kosaki, *Extension of Jones' theory on index to arbitrary factors*, J. Funct. Anal. 66 (1986).

[11] R. Longo, *A duality for Hopf algebras and for subfactors I*, CMP 159 (1994).

[12] J. Cardy, I. Peschel, *Finite-size dependence of the free energy in two-dimensional critical systems*, Nucl. Phys. B 300 (1988).

[13] C. L. Douglas, A. Henriques, *Topological modular forms and conformal nets*, in: Mathematical foundations of quantum field theory and perturbative string theory, Proc. Sympos. Pure Math. 83 (2011), arXiv:1103.4187.